INDIAN INSTITUTE OF TECHNOLOGY MADRAS

MODERN APPLICATION DEVELOPMENT 1 (MAD 1 / APPDEV 1)

Comprehensive Past Year CBT Examination Question Bank (301 Authentic Questions)

Curriculum Week	Topic Description	Questions
Week 1	Internet Architecture, HTTP & Network Calculations	50 Qs
Week 2	HTML5, CSS3, Box Model & Form Controls	115 Qs
Week 3	Backend Python, Flask Essentials, Routing & Request Handling	78 Qs
Week 4	Jinja2 Templating, Dynamic Rendering, SQLite & CRUD Applications	58 Qs
Total	**Verified Authentic Past Exam Questions Across All 4 Weeks**	**301 Questions**

WEEK 1: INTERNET ARCHITECTURE, HTTP & NETWORK CALCULATIONS

- **1.1 Web & Internet Basics (Client-Server Architecture, Protocols & Servers) (10 Questions)**

MAD1_007 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 70 Question Id : 6406531298439 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    p {
      color: blue;
    }
    #highlight {
      color: green;
    }
    .important {
      color: red;
    }
  </style>
</head>
<body>
  <p class="important" id="highlight">This is a paragraph.</p>
</body>
</html>
```

What will be the color of the paragraph text when rendered in a browser?

ptions :
406534366203. ✖ Blue
406534366204. ✔ Green
406534366205. ✖ Red
406534366206. ✖ The browser default (black)

ub-Section Number :	4
ub-Section Id :	640653206740
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406534366204

MAD1_017 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 106 Question Id : 640653345241 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

Which of the following statement is true about MVC architecture?

ptions :

406531148331. ✖ The view redirects the incoming request to model.

Official Answer Key: 6406531148333

MAD1_028 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 105 Question Id : 640653470036 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

Match the component in column A with its correct feature given in column B.

Column A	Column B
1. Telephone Networks	a. Wires occupied only when data to be sent
2. Packet Switched Networks	b. First public packet switched network
3. ARPANET	c. Physical wires tied up for duration of call even if nothing said
4. TCP	d. Establishes reliable communications

ptions :

406531561998. ✖ 1 - a, 2 - b, 3 - c, 4 - d

406531561999. ✖ 1 - b, 2 - d, 3 - a, 4 - c

406531562000. ✖ 1 - d, 2 - c, 3 - b, 4 - a

406531562001. ✔ 1 - c, 2 - a, 3 - b, 4 - d

Official Answer Key: 6406531562001

MAD1_035 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 114 Question Id : 640653470035 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 2

uestion Label : Multiple Choice Question

Match the component in column A with its correct feature given in column B.

Column A	Column B
1. Server	a. Data Pipeline
2. Client	b. Explicit Server(s), Explicit Client(s)
3. Client-Server Architecture	c. User Interaction
4. Network	d. Provides data on demand

ptions :

406531561994. ✖ 1 - a, 2 - b, 3 - c, 4 - d

406531561995. ✖ 1 - b, 2 - d, 3 - a, 4 - c

406531561996. ✔ 1 - d, 2 - c, 3 - b, 4 - a

406531561997. ✖ 1 - c, 2 - a, 3 - b, 4 - d

ub-Section Number :	4
ub-Section Id :	64065367682
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406531561996

MAD1_093 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz1 May23 .pdf

uestion Number : 111 Question Id : 640653577867 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

uestion Label : Multiple Choice Question

/hat is the correct sequence of working of Client-Server Architecture.

. The client asks for the IP address for the particular server from the Domain Name system.
. To that IP address, the client sends the request to the particular server with that port number
hat is specified to the particular application and then server responds
. The DNS server responds with the IP address.
. The response message is received by the client and based on that port number, the response
acket is consumed by the application to which it belongs.

ptions :

406531929703. ✖ c → a → b → d

406531929704. ✖ a → b → c → d

406531929705. ✔ a → c → b → d

406531929706. ✖ None

Official Answer Key: 6406531929705

MAD1_100 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 70 Question Id : 6406531297899 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    p {
      color: blue;
    }
    #highlight {
      color: green;
    }
    .important {
      color: red;
    }
  </style>
</head>
<body>
  <p class="important" id="highlight">This is a paragraph.</p>
</body>
</html>
```

What will be the color of the paragraph text when rendered in a browser?

ptions :
406534364661. ✗ Blue
406534364662. ✓ Green
406534364663. ✗ Red
406534364664. ✗ The browser default (black)

ub-Section Number :	4
ub-Section Id :	640653206538
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406534364662

MAD1_115 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 70 Question Id : 6406531298169 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    p {
      color: blue;
    }
    #highlight {
      color: green;
    }
    .important {
      color: red;
    }
  </style>
</head>
<body>
  <p class="important" id="highlight">This is a paragraph.</p>
</body>
</html>
```

What will be the color of the paragraph text when rendered in a browser?

ptions :
406534365432. ✖ Blue
406534365433. ✔ Green
406534365434. ✖ Red
406534365435. ✖ The browser default (black)

ub-Section Number :	4
ub-Section Id :	640653206639
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406534365433

MAD1_255 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf

uestion Number : 115 Question Id : 640653668552 Question Type : MSQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2 Max. Selectable Options : 0

uestion Label : Multiple Select Question

Which of the following statements regarding dynamic webpages is/are true?

ptions :

406532239847. ✖ The page is already present at the server even before a user requests it.

406532239848. ✖ The page generally maintains the same content every time the user requests

406532239849. ✔ The page is generated on demand by a program or a request from a browse

406532239850. ✔ On every request, every new page created may be different from the last.

ub-Section Number :	6
ub-Section Id :	64065395169

Official Answer Key: 6406532239849, 6406532239850

MAD1_257 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 115 Question Id : 640653668913 Question Type : MSQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2 Max. Selectable Options : 0

uestion Label : Multiple Select Question

Which of the following statements regarding dynamic webpages is/are true?

ptions :

406532240943. ✖ The page is already present at the server even before a user requests it.

406532240944. ✖ The page generally maintains the same content every time the user requests

406532240945. ✔ The page is generated on demand by a program or a request from a browse

406532240946. ✔ On every request, every new page created may be different from the last.

ub-Section Number :	6
ub-Section Id :	64065395274

Official Answer Key: 6406532240945, 6406532240946

MAD1_259 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN4 EXAM QPD4 29 Oct.pdf

uestion Number : 115 Question Id : 640653669174 Question Type : MSQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2 Max. Selectable Options : 0

uestion Label : Multiple Select Question

Which of the following statements regarding dynamic webpages is/are true?

ptions :

406532241695. ✖ The page is already present at the server even before a user requests it.

406532241696. ✖ The page generally maintains the same content every time the user requests

406532241697. ✔ The page is generated on demand by a program or a request from a browse

406532241698. ✔ On every request, every new page created may be different from the last.

ub-Section Number :	6
ub-Section Id :	64065395373

Official Answer Key: 6406532241697, 6406532241698

• 1.2 Data Transmission, Bandwidth, RTT & Propagation Delay Calculations (17 Questions)

MAD1_010 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf, IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 73 Question Id : 6406531298442 Question Type : SA
orrect Marks : 3
uestion Label : Short Answer Question
/e want to use an online grammar checking tool, which takes each word from the client and ends it to the web server. The available network bandwidth between the client and the server is ibps. What should be the maximum allowed size(in MBs) of a document containing the word ser om the client to the server per second for the grammar check? (Use: 1KB= 1000Bytes, MB=1000KB and so on.)
esponse Type : Numeric
valuation Required For SA : Yes
how Word Count : Yes
nswers Type : Equal
ext Areas : PlainText
ossible Answers :
50

uestion Number : 74 Question Id : 6406531298443 Question Type : SA
orrect Marks : 3

Official Answer Key: 250

MAD1_023 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 116 Question Id : 640653345255 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

or a network bandwidth of 8 Gbps, what should be the size of each request if 5000 such request re to be sent over the network per second? [**Use these relations:** 1 Byte = 8 bits, 1 KB = 1000 ytes, 1 MB = 1000 KBs and so on.]

ptions :

406531148387. ✖ 1.6 KB

406531148388. ✔ 200 KB

406531148389. ✖ 1.6 MB

406531148390. ✖ 200 MB

ub-Section Number :	4
ub-Section Id :	64065348774
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406531148388

MAD1_039 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf, IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

uestion Number : 124 Question Id : 640653737406 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

certain network connection with a bandwidth of 12 Mbps allows making 4000 requests to a erver per second. If each request is modified to have an additional data of 6 Kb, what should be he additional bandwidth required to maintain the rate of 4000 requests per second?

ptions :

406532467946. ✖ 12 Mbps

406532467947. ✔ 24 Mbps

406532467948. ✖ 36 Mbps

406532467949. ✖ 48 Mbps

Official Answer Key: 6406532467947

MAD1_056 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 78 Question Id : 6406531113770 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

server S requests to retrieve data from two data centers DC_1 and DC_2, which are ocated at distances of D_1 and D_2 meters respectively from the server S. The speed of lata travelling through the medium that connects S with DC_1 is m_1 m/sec and that onnects S with DC_2 is m_2 m/sec. With the given media for data transfer between S → DC_1 nd S → DC_2, it takes an additional 200 milliseconds for the data to reach S from C_2 than it takes from DC_1 if the complete response is ready only when the data is vailable to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms f D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a equest from the server]

ptions :

406533774184. ✖ $m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$

406533774185. ✖ $m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$

406533774186. ✖ $m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$

406533774187. ✔ $m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$

Official Answer Key: 6406533774187

MAD1_070 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 78 Question Id : 6406531114397 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

server S requests to retrieve data from two data centers DC_1 and DC_2, which are ocated at distances of D_1 and D_2 meters respectively from the server S. The speed of lata travelling through the medium that connects S with DC_1 is m_1 m/sec and that onnects S with DC_2 is m_2 m/sec. With the given media for data transfer between S → DC_1 nd S → DC_2, it takes an additional 200 milliseconds for the data to reach S from C_2 than it takes from DC_1 if the complete response is ready only when the data is vailable to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms f D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a equest from the server]

ptions :

406533775997. ✖ $m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$

406533775998. ✖ $m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$

406533775999. ✖ $m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$

406533776000. ✔ $m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$

Official Answer Key: 6406533776000

MAD1_084 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 78 Question Id : 6406531114110 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

server S requests to retrieve data from two data centers DC_1 and DC_2, which are ocated at distances of D_1 and D_2 meters respectively from the server S. The speed of lata travelling through the medium that connects S with DC_1 is m_1 m/sec and that onnects S with DC_2 is m_2 m/sec. With the given media for data transfer between S → DC_1 nd S → DC_2, it takes an additional 200 milliseconds for the data to reach S from C_2 than it takes from DC_1 if the complete response is ready only when the data is vailable to S from both the datacenters. Figure out the value of m_2 (in m/sec) in terms f D_1, D_2 and m_1. [Note: The data centers send responses only when they receive a equest from the server]

ptions :

406533775168. ✖ $m_2 = \frac{D_2 m_1}{D_1 - 0.2m_1}$

406533775169. ✖ $m_2 = \frac{D_2 m_1}{D_1 + 0.2m_1}$

406533775170. ✖ $m_2 = \frac{2D_2 m_1}{2D_1 - 0.2m_1}$

406533775171. ✔ $m_2 = \frac{2D_2 m_1}{2D_1 + 0.2m_1}$

Official Answer Key: 6406533775171

MAD1_091 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz1 May23 .pdf

uestion Number : 107 Question Id : 640653577855 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

Let L = {'a', 'b', 'c', 'd', 'A', 'B', 'C', 'D', '0', '1', ' '} be a complete character set (i.e., only these characters can be used to represent text in the document). If a document that uses fixed encoding for all characters is created using the character set L and has a disk size of 2 Kilobytes, the number of characters in the documents would be_____. [Take 1 Byte = 8 bits, 1 KB = 1000 Bytes, 1 MB = 1000 Kilobytes and so on.]

ptions :

406531929659. ✖ 2000

406531929660. ✔ 4000

406531929661. ✖ 8000

406531929662. ✖ 16000

Official Answer Key: 6406531929660

MAD1_128 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 38 Question Id : 640653825022 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

certain movie with advertisements is being streamed by the client on the internet with a andwidth of 5 Mbps. If the total data consumed only in the advertisements is 340 MB and the dvertisements comprises 17% of the total duration of the streamed movie, what can be the size f the entire movie (in Megabytes) including the advertisements?

ptions :

406532774735. ✖ 20

406532774736. ✖ 640

406532774737. ✔ 2000

406532774738. ✖ 1660

Official Answer Key: 6406532774737

MAD1_134 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 45 Question Id : 640653825027 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

onsider a client A that makes a request to server B that fetches the information from Datacente which in turn fetches other required information from the Datacenter D. The distance between he client A, server B and datacenters C and D within the network and the speed of information i he medium connecting these nodes is mentioned in the table below.

Nodes	A to B	B to C	C to D
Distance (km)	12000	8500	7500
Speed of Information(m/s)	3×10^8	2×10^8	1.5×10^8

alculate the round trip latency (in milliseconds) of the network?

ptions :

406532774755. ✖ 187

406532774756. ✖ 132.5

406532774757. ✔ 265

406532774758. ✖ 150

Official Answer Key: 6406532774757

MAD1_253 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf

uestion Number : 113 Question Id : 640653668554 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5

uestion Label : Multiple Choice Question

. mobile server and a client initially 750 km apart are moving towards each other in a straight lin
vith the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant t
Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the
erver collide. [speed of light in vacuum: 3×10^8 m/sec, 1 MB = 1000 KB and so on.]

ptions :

406532239855. ✖ 21.6

406532239856. ✖ 6

406532239857. ✔ 2.7

406532239858. ✖ 7.5

ub-Section Number :	5
ub-Section Id :	64065395168
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406532239857

MAD1_256 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 113 Question Id : 640653668915 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5

uestion Label : Multiple Choice Question

. mobile server and a client initially 750 km apart are moving towards each other in a straight lin
vith the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant t
Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the
erver collide. [speed of light in vacuum: 3×10^8 m/sec, 1 MB = 1000 KB and so on.]

ptions :

406532240951. ✖ 21.6

406532240952. ✖ 6

406532240953. ✔ 2.7

406532240954. ✖ 7.5

ub-Section Number :	5
ub-Section Id :	64065395273
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406532240953

MAD1_258 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN4 EXAM QPD4 29 Oct.pdf

uestion Number : 113 Question Id : 640653669176 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5
uestion Label : Multiple Choice Question
. mobile server and a client initially 750 km apart are moving towards each other in a straight lin
vith the speeds of 165 kmph and 85 kmph respectively. The network bandwidth is set constant t
Mbps. Determine how much data (in Gigabytes) is used by the client before the client and the
erver collide. [speed of light in vacuum: 3 *x* 10^8 m/sec, 1 MB = 1000 KB and so on.]
ptions :

406532241703. ✖ 21.6

406532241704. ✖ 6

406532241705. ✔ 2.7

406532241706. ✖ 7.5

ub-Section Number :	5
ub-Section Id :	64065395372
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406532241705

MAD1_260 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf, IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 101 Question Id : 640653995532 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
uppose a Gamer wants to buy a Bluetooth earphone which can give him a good gaming xperience, for that the delay between the action that happens in his phone and the sound hear y him for that action should be as low as possible so that he can play well. Following this cenario, which of the following option(s) is/are correct?
ptions :

406533360884. ✖ He should buy the earphones which have high latency as much possible.

406533360885. ✖ It all depends on Internet Speed, so we can't determine anything.

406533360886. ✔ He should buy the earphones which have low latency as much possible.

406533360887. ✖ He should buy the earphones which have as low Bluetooth version as possibl

Official Answer Key: 6406533360886

MAD1_268 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf, IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 109 Question Id : 640653995539 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

certain network connection with a bandwidth of 10 Mbps allows making 1000 requests to a erver per second. If each request is modified to have an additional data of 8 Kb, what should be ne additional bandwidth required to maintain the rate of 2000 requests per second?

ptions :

406533360908. ✖ 18 Mbps

406533360909. ✖ 16 Mbps

406533360910. ✔ 26 Mbps

406533360911. ✖ 36 Mbps

Official Answer Key: 6406533360910

MAD1_270 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 111 Question Id : 640653995546 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

A client C and the server S located, 12000 km apart are connected with the help of cable. A request was sent by the client C to server S, but while sending back the esponse to the client, the cable breaks and the response is now sent back via air. This hange in medium caused an additional delay of 20 msec. As compared to the Round rip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

ptions :

406533360937. ✖ Increase

406533360938. ✖ Decrease

406533360939. ✔ Remain Same

406533360940. ✖ Insufficient Information

ub-Section Number :	5
ub-Section Id :	640653146813
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533360939

MAD1_284 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 111 Question Id : 640653995828 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

A client C and the server S located, 12000 km apart are connected with the help of cable. A request was sent by the client C to server S, but while sending back the esponse to the client, the cable breaks and the response is now sent back via air. This hange in medium caused an additional delay of 20 msec. As compared to the Round rip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

ptions :

406533361721. ✖ Increase

406533361722. ✖ Decrease

406533361723. ✔ Remain Same

406533361724. ✖ Insufficient Information

ub-Section Number :	5
ub-Section Id :	640653146902
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533361723

MAD1_297 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 111 Question Id : 640653996110 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

A client C and the server S located, 12000 km apart are connected with the help of cable. A request was sent by the client C to server S, but while sending back the esponse to the client, the cable breaks and the response is now sent back via air. This hange in medium caused an additional delay of 20 msec. As compared to the Round rip Time of the healthy network, the Round Trip Time (RTT) of the faulty system would _____.

The speed of light in air is 3×10^8 m/s and that on a cable is 2×10^8 m/s]

ptions :

406533362505. ✖ Increase

406533362506. ✖ Decrease

406533362507. ✔ Remain Same

406533362508. ✖ Insufficient Information

ub-Section Number :	5
ub-Section Id :	640653146991
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533362507

• 1.3 Character Encoding & Representation (ASCII, Unicode, UTF-8 & Binary) (6 Questions)

MAD1_018 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 107 Question Id : 640653345246 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

he hexadecimal representation of the number 5457_8 is _____ .

ptions :

406531148351. ✖ A2F

406531148352. ✖ B1F

406531148353. ✔ B2F

406531148354. ✖ C1E

Official Answer Key: 6406531148353

MAD1_031 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 110 Question Id : 640653470044 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

text document of size 2000 bytes. Each character in the document are 8 bits ASCII encoding.

What is the number of characters in the text document?

Note: 8 bits = 1 byte]

ptions :

406531562030. ✖ 250

406531562031. ✖ 1000

406531562032. ✔ 2000

406531562033. ✖ 8000

ub-Section Number :	3
ub-Section Id :	64065367681
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406531562032

MAD1_057 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 79 Question Id : 6406531113771 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

ptions :

406533774188. ✖ `0000:3B2B:9821:0652`

406533774189. ✖ `3B2B:9821:0652:0000`

406533774190. ✔ `0000:0652:9821:3B2B`

406533774191. ✖ `0652:9821:3B2B:0000`

Official Answer Key: 6406533774190

MAD1_071 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 79 Question Id : 6406531114398 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

ptions :

406533776001. ✖ `0000:3B2B:9821:0652`

406533776002. ✖ `3B2B:9821:0652:0000`

406533776003. ✔ `0000:0652:9821:3B2B`

406533776004. ✖ `0652:9821:3B2B:0000`

Official Answer Key: 6406533776003

MAD1_085 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 79 Question Id : 6406531114111 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
Consider two fictitious addressing systems, IPv5 and IPv8, used to store the IP address of a machine. Following are the forms of storing addresses for each system.

IPv5: It is a 45-bit long address denoted by 5 sets of 3-digit octal numbers separated by a hyphen **(-)**.
e.g. `xxx - xxx - xxx - xxx - xxx`

IPv8: It is a 64-bit long address denoted by 4 sets of 4-bit hexadecimal numbers separated by colon **(:)**
e.g. `xxxx:xxxx:xxxx:xxxx`

How would an IPv5 address, `145-123-010-235-453` be represented in IPv8 format based on information given above?

ptions :

406533775172. ✖ `0000:3B2B:9821:0652`

406533775173. ✖ `3B2B:9821:0652:0000`

406533775174. ✔ `0000:0652:9821:3B2B`

406533775175. ✖ `0652:9821:3B2B:0000`

Official Answer Key: 6406533775174

MAD1_250 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

)uestion Number : 108 Question Id : 640653668553 Question Type : MCQ Is Question
Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

)uestion Label : Multiple Choice Question

onsider a character set L, which consists of only those characters which are used in the tatement, “pack my box with five dozen liquor jugs”. If this statement is to be saved in a ocument with minimum encoding, what will be the size of the document given that no other nformation or context is to be saved?

)ptions :

406532239851. ✖ 160 bits

406532239852. ✖ 320 bits

406532239853. ✔ 195 bits

406532239854. ✖ 390 bits

Official Answer Key: 6406532239853

• 1.4 HTTP Protocol, URLs, Request/Response Headers & Status Codes (17 Questions)

MAD1_032 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 111 Question Id : 640653470031 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

am wants to do the following operations on his bank website through the browser.

ind the HTTP methods conventionally used for the following operations?

. To create a new fixed deposit account

. To read his transaction history

. To update the account after the transaction of money.

. To remove his unused account.

ptions :

406531561978. ✖ 1 - GET, 2- POST, 3 - PUT, 4 - DELETE

406531561979. ✖ 1 - POST, 2- GET, 3 - PUT, 4 - PATCH

406531561980. ✔ 1 - POST, 2- GET, 3 - PUT, 4 - DELETE

406531561981. ✖ 1 - PUT, 2- GET, 3 - POST, 4 - DELETE

Official Answer Key: 6406531561980

MAD1_046 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 67 Question Id : 6406531113759 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which of the following is the correct way of calling an external CSS file using TML?

ptions :

406533774140. ✖
```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

406533774141. ✔
```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

406533774142. ✖
```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Official Answer Key: 6406533774141

MAD1_047 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 68 Question Id : 6406531113760 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which command would you use to get detailed information about the HTTP equest and response, including headers, and to help debug the HTTP request when sing curl?

ptions :

406533774144. ✖ `curl -i <URL>`

406533774145. ✔ `curl -v <URL>`

406533774146. ✖ `curl --headers <URL>`

406533774147. ✖ `curl -X GET <URL>`

Official Answer Key: 6406533774145

MAD1_061 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 67 Question Id : 6406531114386 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which of the following is the correct way of calling an external CSS file using TML?

ptions :

406533775953. ✖
```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

406533775954. ✔
```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

406533775955. ✖
```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Official Answer Key: 6406533775954

MAD1_062 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 68 Question Id : 6406531114387 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which command would you use to get detailed information about the HTTP equest and response, including headers, and to help debug the HTTP request when sing curl?

ptions :

406533775957. ✖ `curl -i <URL>`

406533775958. ✔ `curl -v <URL>`

406533775959. ✖ `curl --headers <URL>`

406533775960. ✖ `curl -X GET <URL>`

Official Answer Key: 6406533775958

MAD1_075 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 67 Question Id : 6406531114099 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which of the following is the correct way of calling an external CSS file using TML?

ptions :

406533775124. ✖
```
<head>
<link url="stylesheet" type="text/css" href="mystyle.css">
</head>
```

406533775125. ✔
```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

406533775126. ✖
```
<head>
<link href="stylesheet" type="text/css" file="mystyle.css">
</head>
```

Official Answer Key: 6406533775125

MAD1_076 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 68 Question Id : 6406531114100 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which command would you use to get detailed information about the HTTP equest and response, including headers, and to help debug the HTTP request when sing curl?

ptions :

406533775128. ✖ `curl -i <URL>`

406533775129. ✔ `curl -v <URL>`

406533775130. ✖ `curl --headers <URL>`

406533775131. ✖ `curl -X GET <URL>`

Official Answer Key: 6406533775129

MAD1_089 | Type: MCQ | Marks: +2.0
Source Exam(s): Quiz1 May23 .pdf

uestion Number : 103 Question Id : 640653577854 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

Which of the following statements is/are true?

ptions :

406531929655. ✖ HTTP headers (for example “Content-Type”) are case-sensitive

406531929656. ✖ The body/payload in an HTTP request is mandatory

406531929657. ✔ The “Content-Length” header indicates the size of the request body in bytes

406531929658. ✖ All of these

Official Answer Key: 6406531929657

MAD1_125 | Type: MCQ | Marks: +2.0
Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 35 Question Id : 640653825021 Question Type : MCQ

orrect Marks : 2

uestion Label : Multiple Choice Question

Which of the below HTTP status codes represents the client error?

ptions :

406532774730. ✖ 500-599

Official Answer Key: 6406532774731

MAD1_248 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 103 Question Id : 640653668543 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 2
uestion Label : Multiple Choice Question
latch the following HTTP methods in Column A with their conventional usage in Column B.

	Column A		Column B
1	GET	a	Send data to a server to create/update a resource
2	HEAD	b	Update a resource
3	PUT	c	method, generally used to request a resource from the server.
4	POST	d	Asks for a response identical to that of a GET request, but without the response body

ptions :

406532239811. ✔ 1-c, 2-d, 3-b, 4-a

406532239812. ✘ 1-c, 2-b, 3-d, 4-a

406532239813. ✘ 1-c, 2-d, 3-a, 4-b

406532239814. ✘ 1-a, 2-d, 3-b, 4-c

ub-Section Number :	3
ub-Section Id :	64065395166
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406532239811

MAD1_261 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 102 Question Id : 640653995543 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

onsider the below URL:
ttps://www.google.com/search?q=physics+books

elect the correct answer that identifies the different components of the given URL.

ptions :

406533360925. ✖
Host: http
Protocol: google.com/search
Query string: q=physics+books

406533360926. ✔
Protocol: http
Host: google.com/search
Query string: q=physics+books

406533360927. ✖
Protocol: http
Query string: google.com/search
Protocol: q=physics+books

406533360928. ✖
Host: http
Query string: google.com/search
Protocol: q=physics+books

ub-Section Number :	3
ub-Section Id :	640653146811
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533360926

MAD1_265 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 106 Question Id : 640653995540 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he directory listing of a folder and the content of each of its files are given below.

irectory Listing:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

ilename: base.html

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

ilename: home.html

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

ilename: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

f a simple Python HTTP server is created for this directory, what will be rendered on the rowser for base URL?

ptions :

406533360912. ✗ **Welcome to PDSA course**

406533360913. ✓ **Welcome to DBMS course**

406533360914. ✗ **Welcome to MAD-I course**

406533360915. ✗ **Directory listing for /**

Official Answer Key: 6406533360913

MAD1_271 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf, IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 112 Question Id : 640653995542 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the below server response

HTTP/2.0 500 Internal Server Error

Which of the following is/are **True**?

ptions :

406533360920. ✔ HTTP/2.0 is the protocol and version respectively

406533360921. ✖ 500 is the port number

406533360922. ✔ "Internal Server Error" is the status message

406533360923. ✔ 500 is the status code

406533360924. ✖ HTTP/2.0 is the server failure

Official Answer Key: 6406533360920, 6406533360922, 6406533360923

MAD1_276 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 102 Question Id : 640653995825 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

onsider the below URL:
ttps://www.google.com/search?q=physics+books

elect the correct answer that identifies the different components of the given URL.

ptions :

406533361709. ✖
Host: http
Protocol: google.com/search
Query string: q=physics+books

406533361710. ✔
Protocol: http
Host: google.com/search
Query string: q=physics+books

406533361711. ✖
Protocol: http
Query string: google.com/search
Protocol: q=physics+books

406533361712. ✖
Host: http
Query string: google.com/search
Protocol: q=physics+books

ub-Section Number :	3
ub-Section Id :	640653146900
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533361710

MAD1_280 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 106 Question Id : 640653995822 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he directory listing of a folder and the content of each of its files are given below.

irectory Listing:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

ilename: base.html

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

ilename: home.html

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

ilename: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

f a simple Python HTTP server is created for this directory, what will be rendered on the rowser for base URL?

ptions :

406533361696. ✗ **Welcome to PDSA course**

406533361697. ✓ **Welcome to DBMS course**

406533361698. ✗ **Welcome to MAD-I course**

406533361699. ✗ **Directory listing for /**

Official Answer Key: 6406533361697

MAD1_289 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 102 Question Id : 640653996107 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

onsider the below URL:
ttps://www.google.com/search?q=physics+books

elect the correct answer that identifies the different components of the given URL.

ptions :

406533362493. ✖
Host: http
Protocol: google.com/search
Query string: q=physics+books

406533362494. ✔
Protocol: http
Host: google.com/search
Query string: q=physics+books

406533362495. ✖
Protocol: http
Query string: google.com/search
Protocol: q=physics+books

406533362496. ✖
Host: http
Query string: google.com/search
Protocol: q=physics+books

ub-Section Number :	3
ub-Section Id :	640653146989
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533362494

MAD1_293 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 106 Question Id : 640653996104 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he directory listing of a folder and the content of each of its files are given below.

irectory Listing:

```
my_dir-|
       |-- base.html
       |-- home.html
       |-- index.html
```

ilename: base.html

```
<body>
    <h1>Welcome to MAD-I course</h1>
</body>
```

ilename: home.html

```
<body>
    <h1>Welcome to PDSA course</h1>
</body>
```

ilename: index.html

```
<body>
    <h1>Welcome to DBMS course</h1>
</body>
```

f a simple Python HTTP server is created for this directory, what will be rendered on the rowser for base URL?

ptions :

406533362480. ✗ **Welcome to PDSA course**

406533362481. ✓ **Welcome to DBMS course**

406533362482. ✗ **Welcome to MAD-I course**

406533362483. ✗ **Directory listing for /**

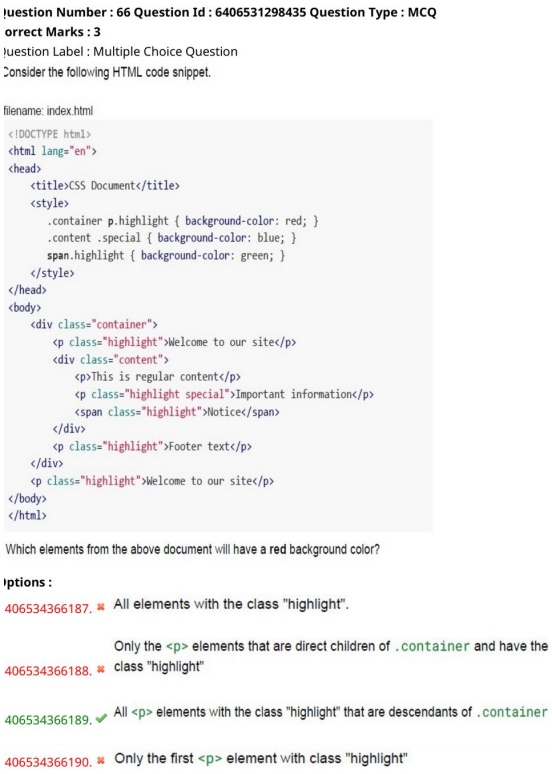
Official Answer Key: 6406533362481

WEEK 2: HTML5, CSS3, BOX MODEL & FORM CONTROLS

• 2.1 HTML5 Elements, Semantic Tags & Document Structure (30 Questions)

MAD1_003 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf



uestion Number : 66 Question Id : 6406531298435 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider the following HTML code snippet.

filename: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>CSS Document</title>
    <style>
       .container p.highlight { background-color: red; }
       .content .special { background-color: blue; }
       span.highlight { background-color: green; }
    </style>
</head>
<body>
    <div class="container">
        <p class="highlight">Welcome to our site</p>
        <div class="content">
            <p>This is regular content</p>
            <p class="highlight special">Important information</p>
            <span class="highlight">Notice</span>
        </div>
        <p class="highlight">Footer text</p>
    </div>
    <p class="highlight">Welcome to our site</p>
</body>
</html>
```

Which elements from the above document will have a **red** background color?

ptions :

406534366187. ✗ All elements with the class "highlight".

406534366188. ✗ Only the <p> elements that are direct children of .container and have the class "highlight"

406534366189. ✓ All <p> elements with the class "highlight" that are descendants of .container

406534366190. ✗ Only the first <p> element with class "highlight"

Official Answer Key: 6406534366189

MAD1_019 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 108 Question Id : 640653345250 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

onsider a file consisting of 2000 alphanumeric characters including spaces. How much would be

ne total occupied space in bits, assuming UCS-4 is used?

ptions :

406531148367. ✔ 64000 bits

406531148368. ✖ 32000 bits

406531148369. ✖ 24000 bits

406531148370. ✖ 12000 bits

Official Answer Key: 6406531148367

MAD1_020 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 110 Question Id : 640653345242 Question Type : MCQ Is Question
andatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

uestion Label : Multiple Choice Question

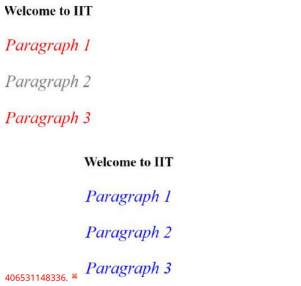
onsider the following HTML code below.

```
!DOCTYPE html>
html lang="en">
  <head>
    <meta charset="UTF-8"/>
    <style>
      body{text-align: center}
      p{font-size: 30px;font-style: italic;color: black;}
      .blue{color: green;}
      .red{color: blue;}
      .green{color: red;}
      #myId{color: grey;}
    </style>
  </head>
  <body>
    <div>
      <h2>Welcome to IIT</h2>
      <p class="blue red green">Paragraph 1</p>
      <p class="red blue green" id="myId">Paragraph 2</p>
      <p class="green red blue">Paragraph 3 </p>
    </div>
  </body>
/html>
```

ow will the browser render the above given HTML file?

ptions :

406531148335. ✔



406531148336. ✖

Official Answer Key: 6406531148335

MAD1_021 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf, Quiz1 May23 .pdf

uestion Number : 111 Question Id : 640653345244 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

onsider the following Python code snippet.

ilename: code.py

```
import sys
my_args = sys.argv

print("The python file name is:", my_args[1])
print(f"My username is: {my_args[3]}{my_args[4]}")
```

What will be the output on console if the command given is:

```
python code.py output.py Appdev Michael 2003 1003
```

ptions :

406531148343. ✖
```
The python file name is: code.py
My username is: Michael1003
```

406531148344. ✖
```
The python file name is: code.py
My username is: Michael2003
```

406531148345. ✖
```
The python file name is: output.py
My username is: Michael1003
```

406531148346. ✔
```
The python file name is: output.py
My username is: Michael2003
```

Official Answer Key: 6406531148346

MAD1_022 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf, Quiz 1 Jan23 .pdf

uestion Number : 113 Question Id : 640653345247 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

nnie and Curly took Database and Networks courses. Bella took Compiler and Python ourses and Deva took Compiler and Database courses. Which of the following list of uples correctly represents the relationship between names and courses.
ames = {0: 'Annie', 1: 'Bella', 2: 'Curly', 3: 'Deva'}
ourses = {0: 'Database', 1: 'Compiler', 2: 'Networks', 3: 'Python'}

ptions :

406531148355. ✔ rels = [(names[0], courses[0]), (names[2], courses[0]), (names[0], courses[2]), (names[2], courses[2]), (names[3], courses[0]), (names[3], courses[1]), (names[1], courses[1]), (names[1], courses[3])]

406531148356. ✖ rels = [(names[2], courses[0]), (names[3], courses[0]), (names[2], courses[2]), (names[3], courses[2]), (names[0], courses[0]), (names[0], courses[1]), (names[2], courses[1]), (names[2], courses[3])]

406531148357. ✖ rels = [(0, 1), (2,0), (0,2), (2,2), (3,0), (2,1), (1,1), (1,3)]

406531148358. ✖ rels = [(1, 1), (2,2), (0,0), (2,3), (3,1), (2,4), (4,4), (1,4)]

Official Answer Key: 6406531148355

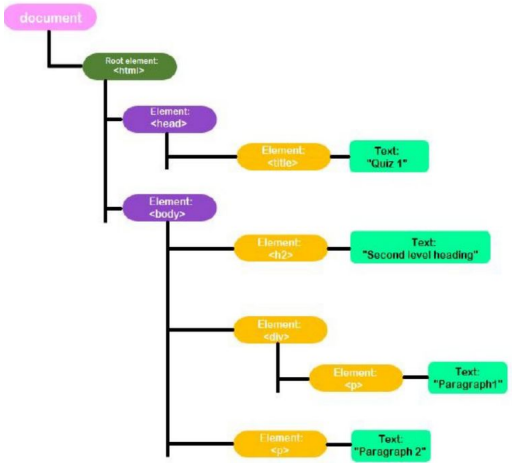
MAD1_026 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 120 Question Id : 640653345254 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5

uestion Label : Multiple Choice Question

onsider the DOM structure given below.



Official Answer Key: 6406531148386

MAD1_027 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 121 Question Id : 640653345256 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5

uestion Label : Multiple Choice Question

onsider a client which is located 6000 kilometers from the server makes a request through the
able. Suddenly after the request reaches the server, the cable breaks and the response is now t
e sent to the client via air with the help of repeaters which added a delay of 75 milliseconds. Ho

ng will the client have to wait before receiving the response? [**Note:** the speed of light on cable
$\times 10^8$ m/sec and that in air is 3×10^8 m/sec.]

ptions :

406531148391. ✗ 50 milliseconds

406531148392. ✗ 135 milliseconds

406531148393. ✓ 125 milliseconds

406531148394. ✗ 115 milliseconds

MLF

ection Id :	64065321859
ection Number :	8
ection type :	Online
landatory or Optional :	Mandatory
umber of Questions :	15
umber of Questions to be attempted :	15
ection Marks :	50
isplay Number Panel :	Yes
roup All Questions :	No
nable Mark as Answered Mark for Review and lear Response :	Yes
laximum Instruction Time :	0
ub-Section Number :	1
ub-Section Id :	64065348776
uestion Shuffling Allowed :	No

Official Answer Key: 6406531148393

MAD1_029 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 106 Question Id : 640653470038 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

/hich of the following is an incorrect way of creating an ordered list in HTML?

ptions :

406531562006. ✗

```
<ol type = "1">
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ol>
```

406531562007. ✗

```
<ol type = "a">
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ol>
```

406531562008. ✓

```
<ul type = "i">
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ul>
```

406531562009. ✗

```
<ol>
      <li>programming</li>
      <li>Data Science</li>
      <li>Artificial Intelligence</li>
  </ol>
```

Official Answer Key: 6406531562008

MAD1_030 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz 1 Jan23 .pdf

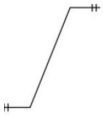
uestion Number : 109 Question Id : 640653470043 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

uestion Label : Multiple Choice Question

Which of the following option represents a one-to-many relationship in the Entity-Relationship iagram?

ptions :

406531562026. ✗



406531562027. ✗

406531562028. ✗

406531562029. ✓

Official Answer Key: 6406531562029

MAD1_033 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 112 Question Id : 640653470033 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

low to represent the number 111_{10} in the binary format?

ptions :

406531561986. ✖ 1110000

406531561987. ✔ 1101111

406531561988. ✖ 1100000

406531561989. ✖ 1111000

Official Answer Key: 6406531561987

MAD1_038 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf, IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

uestion Number : 119 Question Id : 640653737396 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

Consider the following example.

```
<img src="example.jpg" alt=" Gajendra Circle, IITM Campus">
```

Which of the following accessibility principle is followed in this example?

ptions :

406532467910. ✔ Perceivable

406532467911. ✖ Operable

406532467912. ✖ Understandable

406532467913. ✖ Robust

Official Answer Key: 6406532467910

MAD1_040 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf

uestion Number : 131 Question Id : 640653737402 Question Type : MSQ Is Question
andatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
onsider the following statements and choose the correct options.

tatement I: An author can write multiple books, but each book is written by only one author.
tatement II: A publisher can publish multiple books, but each book is published by only one ublisher.
tatement III: A book belongs to one genre, but a genre can have multiple books.
ptions :

406532467934. ✖ Statement I represents a many-to-many relationship between Author and ook.

406532467935. ✔ Statement I represents a one-to-many relationship between Author and Book

406532467936. ✔ Statement II represents a one-to-many relationship between Publisher and ook.

406532467937. ✔ Statement III represents a many-to-one relationship between Book and Genr

ub-Section Number :	7
ub-Section Id :	640653107629
uestion Shuffling Allowed :	No
Section Default? :	null

uestion Id : 640653737403 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
alculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

uestion Numbers : (132 to 133)
uestion Label : Comprehension
nswer the given subquestion.
ub questions

Official Answer Key: 6406532467935, 6406532467936, 6406532467937

MAD1_042 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

uestion Number : 131 Question Id : 640653737688 Question Type : MSQ Is Question
andatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
onsider the following statements and choose the correct options.

tatement I: An author can write multiple books, but each book is written by only one author.
tatement II: A publisher can publish multiple books, but each book is published by only one ublisher.
tatement III: A book belongs to one genre, but a genre can have multiple books.
ptions :

406532468758. ✘ Statement I represents a many-to-many relationship between Author and ook.

406532468759. ✔ Statement I represents a one-to-many relationship between Author and Book

406532468760. ✔ Statement II represents a one-to-many relationship between Publisher and ook.

406532468761. ✔ Statement III represents a many-to-one relationship between Book and Genr

ub-Section Number :	7
ub-Section Id :	640653107742
uestion Shuffling Allowed :	No
Section Default? :	null

uestion Id : 640653737689 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
alculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

uestion Numbers : (132 to 133)
uestion Label : Comprehension
nswer the given subquestion.
ub questions

Official Answer Key: 6406532468759, 6406532468760, 6406532468761

MAD1_044 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024.pdf

uestion Number : 131 Question Id : 640653737974 Question Type : MSQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3 Max. Selectable Options : 0

uestion Label : Multiple Select Question

onsider the following statements and choose the correct options.

tatement I: An author can write multiple books, but each book is written by only one author.
tatement II: A publisher can publish multiple books, but each book is published by only one ublisher.
tatement III: A book belongs to one genre, but a genre can have multiple books.
ptions :

406532469582. ✖ Statement I represents a many-to-many relationship between Author and ook.

406532469583. ✔ Statement I represents a one-to-many relationship between Author and Book

406532469584. ✔ Statement II represents a one-to-many relationship between Publisher and ook.

406532469585. ✔ Statement III represents a many-to-one relationship between Book and Genr

ub-Section Number :	7
ub-Section Id :	640653107855
uestion Shuffling Allowed :	No
Section Default? :	null

uestion Id : 640653737975 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
alculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

uestion Numbers : (132 to 133)

uestion Label : Comprehension

nswer the given subquestion.

ub questions

Official Answer Key: 6406532469583, 6406532469584, 6406532469585

MAD1_050 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 72 Question Id : 6406531113764 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider the following HTML code:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <title>Document</title>
   <style>
       div {
           color: blue;
       }
       #s1 {
           color: yellow;
       }
       .s2{
           color: green;
       }
       h3{
           color: red;
       }
   </style>
</head>
<body>
   <div class="s2" id="s1"><h3>IITM Online BS Degree Program</h3></div>
</body>
</html>
```

What is the colour of the printed text "IITM Online BS Degree Program"?

ptions :

406533774160. ✖ BLUE
406533774161. ✖ YELLOW
406533774162. ✖ GREEN
406533774163. ✔ RED

Official Answer Key: 6406533774163

MAD1_058 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 80 Question Id : 6406531113772 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
import pyhtml as html

table = {'table': [[1,2,3],[4,5,6]]}

def f_table(ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

Official Answer Key: 6406533774192

MAD1_072 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 80 Question Id : 6406531114399 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
import pyhtml as html

table = {'table': [[1,2,3],[4,5,6]]}

def f_table(ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

Official Answer Key: 6406533776005

MAD1_086 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 80 Question Id : 6406531114112 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
import pyhtml as html

table = {'table': [[1,2,3],[4,5,6]]}

def f_table(ctx):
  template_str = html.html(
    html.head(
      html.title('Title')
    ),
    html.body(
      html.table(
        (html.tr(
            [html.td(cell) for cell in row]
          ) for row in ctx['table'])
      )
    )
  )
  return (template_str)

template = f_table(table)
print(template.render())
```

Official Answer Key: 6406533775176

MAD1_096 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 66 Question Id : 6406531297895 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider the following HTML code snippet.

filename: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>CSS Document</title>
    <style>
       .container p.highlight { background-color: red; }
       .content .special { background-color: blue; }
       span.highlight { background-color: green; }
    </style>
</head>
<body>
    <div class="container">
        <p class="highlight">Welcome to our site</p>
        <div class="content">
            <p>This is regular content</p>
            <p class="highlight special">Important information</p>
            <span class="highlight">Notice</span>
        </div>
        <p class="highlight">Footer text</p>
    </div>
    <p class="highlight">Welcome to our site</p>
</body>
</html>
```

Which elements from the above document will have a **red** background color?

ptions :

406534364645. ✗ All elements with the class "highlight".

406534364646. ✗ Only the <p> elements that are direct children of .container and have the class "highlight"

406534364647. ✓ All <p> elements with the class "highlight" that are descendants of .container

406534364648. ✗ Only the first <p> element with class "highlight"

Official Answer Key: 6406534364647

MAD1_111 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 66 Question Id : 6406531298165 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider the following HTML code snippet.

filename: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>CSS Document</title>
    <style>
       .container p.highlight { background-color: red; }
       .content .special { background-color: blue; }
       span.highlight { background-color: green; }
    </style>
</head>
<body>
    <div class="container">
        <p class="highlight">Welcome to our site</p>
        <div class="content">
            <p>This is regular content</p>
            <p class="highlight special">Important information</p>
            <span class="highlight">Notice</span>
        </div>
        <p class="highlight">Footer text</p>
    </div>
    <p class="highlight">Welcome to our site</p>
</body>
</html>
```

Which elements from the above document will have a **red** background color?

ptions :

406534365416. ✖ All elements with the class "highlight".

406534365417. ✖ Only the <p> elements that are direct children of .container and have the class "highlight"

406534365418. ✔ All <p> elements with the class "highlight" that are descendants of .container

406534365419. ✖ Only the first <p> element with class "highlight"

Official Answer Key: 6406534365418

MAD1_124 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 34 Question Id : 640653825020 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Which of the following is not a correct perceivable accessibility principle?
ptions :
406532774726. ✖ Provide text alternatives for non-text content.
406532774727. ✖ Provide captions and other alternatives for multimedia.
406532774728. ✖ Make it easier for users to see and hear content.
406532774729. ✔ Give users enough time to read and use content.

Official Answer Key: 6406532774729

MAD1_129 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf

uestion Number : 39 Question Id : 640653825024 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

ptions :

Official Answer Key: 6406532774744

MAD1_130 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 40 Question Id : 640653825025 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
he following Python code snippet generates the output on the terminal.

```
rom string import Template
 = Template('$name studies at $university')
 = {'name': 'Bob'}

ut1 = Template('$name studies $subject').safe_substitute(b)
rint(out1)

ut2 = s.substitute(name='Alice', university='Oxford')
rint(out2)

ut3 = Template('$name studies $subject').substitute(b)
rint(out3)
```

hich of the following is the correct output?
ptions :

406532774747. ✖
```
Bob studies at $subject
Alice studies at Oxford
KeyError: 'subject'
```

406532774748. ✔
```
Bob studies $subject
Alice studies at Oxford
KeyError: 'subject'
```

406532774749. ✖
```
Bob studies $subject
Alice studies at Oxford
Bob studies $subject
```

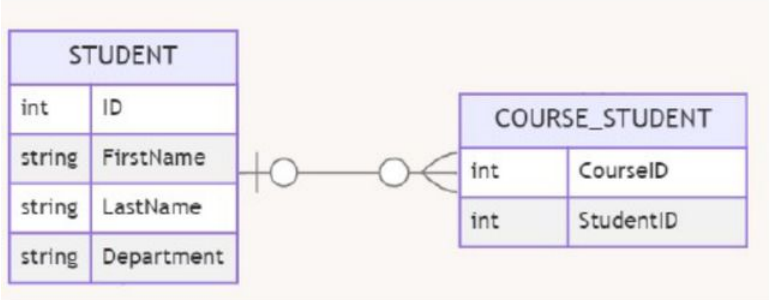
406532774750. ✖
```
Bob studies
Alice studies at Oxford
KeyError: 'subject'
```

Official Answer Key: 6406532774748

MAD1_136 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 47 Question Id : 640653825029 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
/hat can be inferred from the Entity-Relationship Diagram below:



ptions :

406532774763. ✔ A student can exist without enrolling to any course

406532774764. ✔ A student can have more than one courses

406532774765. ✖ A student needs to have at least one course

406532774766. ✖ A course must belong to one and only one student

Official Answer Key: 6406532774763, 6406532774764

MAD1_138 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 39 Question Id : 640653827493 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

ptions :

Official Answer Key: 6406532781268

MAD1_140 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 7 July 2024.pdf

uestion Number : 39 Question Id : 640653827781 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider a document of 1000 characters written using only the characters of the character set L1 = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j'}. If the character set L1 is to be upgraded by adding all the characters of character set L2 = {'0', '1', '2', '3', '4', '5', '6'} in it, By how many bits will the size of the original document increase?

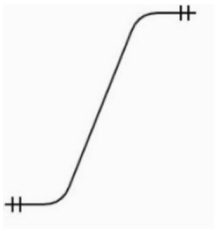
ptions :

Official Answer Key: 6406532782064

MAD1_254 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 114 Question Id : 640653668544 Question Type : MSQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following entities and their relationship has been correctly represented by the ER iagram shown in the image?



ptions :

406532239815. ✔ Person - Passport

406532239816. ✖ Person - Mobile number

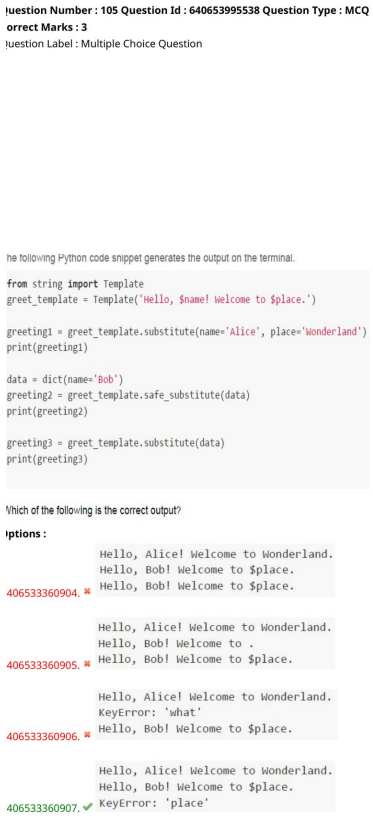
406532239817. ✔ Person - Birth certificate

406532239818. ✔ Person - Fingerprint

Official Answer Key: 6406532239815, 6406532239817, 6406532239818

MAD1_264 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf



uestion Number : 105 Question Id : 640653995538 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he following Python code snippet generates the output on the terminal.

```
from string import Template
greet_template = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)

data = dict(name='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)

greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of the following is the correct output?

ptions :

406533360904. ✗
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

406533360905. ✗
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

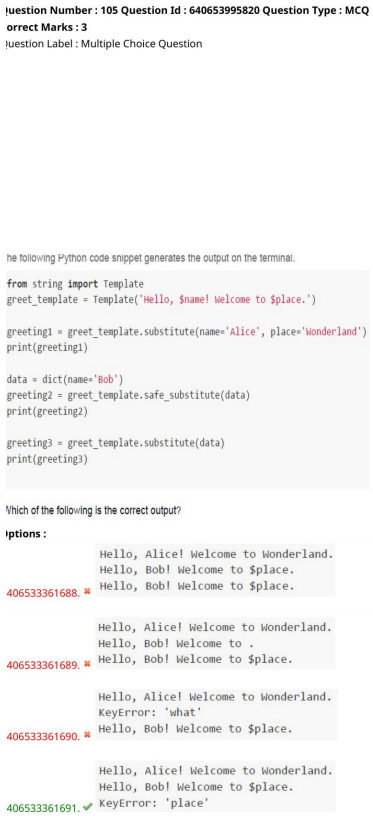
406533360906. ✗
```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

406533360907. ✓
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Official Answer Key: 6406533360907

MAD1_279 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf



uestion Number : 105 Question Id : 640653995820 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he following Python code snippet generates the output on the terminal.

```
from string import Template
greet_template = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)

data = dict(name='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)

greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of the following is the correct output?

ptions :

406533361688. ✗
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

406533361689. ✗
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

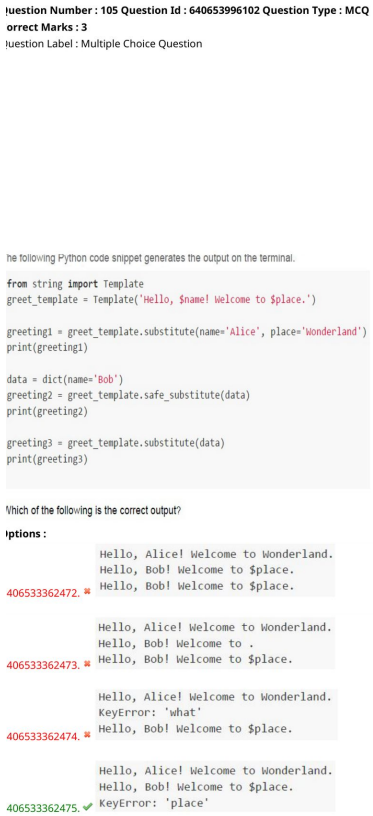
406533361690. ✗
```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

406533361691. ✓
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Official Answer Key: 6406533361691

MAD1_292 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf



uestion Number : 105 Question Id : 640653996102 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he following Python code snippet generates the output on the terminal.

```
from string import Template
greet_template = Template('Hello, $name! Welcome to $place.')

greeting1 = greet_template.substitute(name='Alice', place='Wonderland')
print(greeting1)

data = dict(name='Bob')
greeting2 = greet_template.safe_substitute(data)
print(greeting2)

greeting3 = greet_template.substitute(data)
print(greeting3)
```

Which of the following is the correct output?

ptions :

406533362472. ✗
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
Hello, Bob! Welcome to $place.
```

406533362473. ✗
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to .
Hello, Bob! Welcome to $place.
```

406533362474. ✗
```
Hello, Alice! Welcome to Wonderland.
KeyError: 'what'
Hello, Bob! Welcome to $place.
```

406533362475. ✓
```
Hello, Alice! Welcome to Wonderland.
Hello, Bob! Welcome to $place.
KeyError: 'place'
```

Official Answer Key: 6406533362475

• 2.2 HTML Forms, Input Types, Attributes & Form Submission (64 Questions)

MAD1_012 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 75 Question Id : 6406531298445 Question Type : MSQ
orrect Marks : 4.5 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Which command line input will
roduce the output:

```
flask is a server-side python framework.?
```

ptions :

406534366217. ✖ `python webapp.py flask python type`

406534366218. ✔ `python webapp.py flask type server-side`

406534366219. ✔ `python webapp.py flask type`

406534366220. ✖ `python webapp.py type flask python`

406534366221. ✖ `python webapp.py flask server-side`

Official Answer Key: 6406534366218, 6406534366219

MAD1_024 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 117 Question Id : 640653345243 Question Type : MSQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Select Question

Which of the following statements is/are true about an HTML 5 document?

ptions :

406531148339. ✔ <!DOCTYPE> declaration represents that a given HTML file is HTML5 compliant.

406531148340. ✖ We cannot create HTML pages without <head> and <body> tags.

406531148341. ✔ It is possible to render HTML files without .html extension.

406531148342. ✖ An HTML file will not be rendered if any of its non-self-closing tag is left open.

ub-Section Number :	5
ub-Section Id :	64065348775
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406531148339, 6406531148341

MAD1_054 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 76 Question Id : 6406531113768 Question Type : MSQ

orrect Marks : 3 Max. Selectable Options : 0

uestion Label : Multiple Select Question

ased on the provided latency and response size information, which of the llowing statements is/are correct?

ssume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

ptions :

406533774176. ✔ The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

406533774177. ✔ For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

406533774178. ✖ The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

406533774179. ✔ If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Official Answer Key: 6406533774176, 6406533774177, 6406533774179

MAD1_055 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 77 Question Id : 6406531113769 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

onsider the following two tables in an SQLite database:

Table: `Gifts`

GiftID	GiftName	Price
1	Teddy Bear	250
2	Coffee Mug	150
3	Keychain	100
4	Notebook	180
5	Wall Clock	300

Table: `Reviews`

ReviewID	GiftID	Rating
1	1	4.2
2	2	4.5
3	3	3.9

Official Answer Key: 6406533774181, 6406533774183

MAD1_068 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 76 Question Id : 6406531114395 Question Type : MSQ

orrect Marks : 3 Max. Selectable Options : 0

uestion Label : Multiple Select Question

ased on the provided latency and response size information, which of the llowing statements is/are correct?

ssume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

ptions :

406533775989. ✔ The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

406533775990. ✔ For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

406533775991. ✖ The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

406533775992. ✔ If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Official Answer Key: 6406533775989, 6406533775990, 6406533775992

MAD1_069 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 77 Question Id : 6406531114396 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

onsider the following two tables in an SQLite database:

Table: `Gifts`

GiftID	GiftName	Price
1	Teddy Bear	250
2	Coffee Mug	150
3	Keychain	100
4	Notebook	180
5	Wall Clock	300

Table: `Reviews`

ReviewID	GiftID	Rating
1	1	4.2
2	2	4.5
3	3	3.9

Official Answer Key: 6406533775994, 6406533775996

MAD1_082 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 76 Question Id : 6406531114108 Question Type : MSQ

orrect Marks : 3 Max. Selectable Options : 0

uestion Label : Multiple Select Question

ased on the provided latency and response size information, which of the llowing statements is/are correct?

ssume the speed of light in cable as `2×10^8 m/s` for all the options and `ms` is a millisecond)

ptions :

406533775160. ✔ The speed of light in a cable is approximately `2×10^8 m/s`, resulting in a latency of approximately `5 ms`(millisecond) for a 1000 km distance.

406533775161. ✔ For a data centre located 2000 km away, a one-way request will take approximately `10 ms`, to reach the server, assuming the speed of light in the cable.

406533775162. ✖ The round-trip latency for communication with a data centre 2000 km away is approximately `50ms`.

406533775163. ✔ If a response size is `1 KB`, a maximum of 50 requests per second can be handled considering a round-trip latency of `20ms`.

Official Answer Key: 6406533775160, 6406533775161, 6406533775163

MAD1_083 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 77 Question Id : 6406531114109 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

onsider the following two tables in an SQLite database:

Table: `Gifts`

GiftID	GiftName	Price
1	Teddy Bear	250
2	Coffee Mug	150
3	Keychain	100
4	Notebook	180
5	Wall Clock	300

Table: `Reviews`

ReviewID	GiftID	Rating
1	1	4.2
2	2	4.5
3	3	3.9

Official Answer Key: 6406533775165, 6406533775167

MAD1_104 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 75 Question Id : 6406531297905 Question Type : MSQ
orrect Marks : 4.5 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Which command line input will
roduce the output:

```
flask is a server-side python framework.?
```

ptions :

406534364675. ✖ `python webapp.py flask python type`

406534364676. ✔ `python webapp.py flask type server-side`

406534364677. ✔ `python webapp.py flask type`

406534364678. ✖ `python webapp.py type flask python`

406534364679. ✖ `python webapp.py flask server-side`

Official Answer Key: 6406534364676, 6406534364677

MAD1_119 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 75 Question Id : 6406531298175 Question Type : MSQ
orrect Marks : 4.5 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Which command line input will
roduce the output:

```
flask is a server-side python framework.?
```

ptions :

406534365446. ✖ `python webapp.py flask python type`

406534365447. ✔ `python webapp.py flask type server-side`

406534365448. ✔ `python webapp.py flask type`

406534365449. ✖ `python webapp.py type flask python`

406534365450. ✖ `python webapp.py flask server-side`

Official Answer Key: 6406534365447, 6406534365448

MAD1_137 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 48 Question Id : 640653825033 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
onsider the following Flask code.

ıe name: sum_args.py

```
import sys
from flask import Flask
app = Flask(__name__)

def do_something():
    if len(sys.argv) >= 2:
        l = []
        for i in range(1, len(sys.argv)):
            l.append(int(sys.argv[i]))
        return l
    else:
        return None

@app.route("/")
def do_sum():
    result = do_something()
    if result:
        s = 0
        for e in result:
            s += int(e)
        return f"<h2>Sum is : {s}"
    else:
        return f"<h2>Sum is : {result}"

app.run(debug=True)
```

elect the command(s) to execute the above code without any error.
ptions :

406532774779. ✔ python sum_args.py 2 8 9

406532774780. ✔ python sum_args.py 2

406532774781. ✔ python sum_args.py

406532774782. ✖ python sum_args.py IITM

ub-Section Number :	6
ub-Section Id :	640653122724
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406532774779, 6406532774780, 6406532774781

MAD1_143 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 80 Question Id : 6406531527807 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

onsider the following statements and select the correct option.

tatement 1: The `<span>` elements will start on new lines, while the `<div>` elements will appear inline with surrounding content.

tatement 2: The `<p>` and `<div>` elements will each start on new lines and take up the full width of their container, while `<span>`, `<strong>`, and `<em>` will flow inline with surrounding content.

ptions :

406535115555. ✖ Statement 1 is correct, but statement 2 is incorrect.

406535115556. ✔ Statement 2 is correct, but statement 1 is incorrect.

406535115557. ✖ Both statements 1 and 2 are correct

406535115558. ✖ Both statements 1 and 2 are incorrect

Official Answer Key: 6406535115556

MAD1_147 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 84 Question Id : 6406531527808 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535115559. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535115560. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535115561. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535115562. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535115563. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535115559, 6406535115561, 6406535115562

MAD1_148 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 85 Question Id : 6406531527816 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535115592. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535115593. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535115594. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535115595. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653244911
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535115592, 6406535115594

MAD1_151 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 88 Question Id : 6406531527814 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535115584. ✔ http is the protocol.

406535115585. ✔ 5000 is the port number.

406535115586. ✔ The trailing / represents the base URL .

406535115587. ✖ 5000 is the status code.

Official Answer Key: 6406535115584, 6406535115585, 6406535115586

MAD1_152 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 89 Question Id : 6406531527815 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

img src="logo.png" alt="Company Logo" />

Which of the following is **true**?

ptions :

406535115588. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535115589. ✔ If the image file is missing, browsers display it as a broken image.

406535115590. ✖ The ***alt*** attribute is mandatory in HTML5.

406535115591. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653244913
uestion Shuffling Allowed :	No

uestion Id : 6406531527817 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (90 to 91)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535115588, 6406535115589

MAD1_160 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 84 Question Id : 6406531528111 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535116402. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535116403. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535116404. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535116405. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535116406. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535116402, 6406535116404, 6406535116405

MAD1_161 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 85 Question Id : 6406531528119 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535116435. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535116436. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535116437. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535116438. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653245007
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535116435, 6406535116437

MAD1_164 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 88 Question Id : 6406531528117 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535116427. ✔ http is the protocol.

406535116428. ✔ 5000 is the port number.

406535116429. ✔ The trailing / represents the base URL .

406535116430. ✖ 5000 is the status code.

Official Answer Key: 6406535116427, 6406535116428, 6406535116429

MAD1_165 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 89 Question Id : 6406531528118 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535116431. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535116432. ✔ If the image file is missing, browsers display it as a broken image.

406535116433. ✖ The ***alt*** attribute is mandatory in HTML5.

406535116434. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653245009
uestion Shuffling Allowed :	No

uestion Id : 6406531528120 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (90 to 91)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535116431, 6406535116432

MAD1_171 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 84 Question Id : 6406531528773 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535118333. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535118334. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535118335. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535118336. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535118337. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535118333, 6406535118335, 6406535118336

MAD1_172 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 85 Question Id : 6406531528781 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535118366. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535118367. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535118368. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535118369. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653245192
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535118366, 6406535118368

MAD1_175 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 88 Question Id : 6406531528779 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535118358. ✔ http is the protocol.

406535118359. ✔ 5000 is the port number.

406535118360. ✔ The trailing / represents the base URL .

406535118361. ✖ 5000 is the status code.

Official Answer Key: 6406535118358, 6406535118359, 6406535118360

MAD1_176 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 89 Question Id : 6406531528780 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535118362. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535118363. ✔ If the image file is missing, browsers display it as a broken image.

406535118364. ✖ The ***alt*** attribute is mandatory in HTML5.

406535118365. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653245194
uestion Shuffling Allowed :	No

uestion Id : 6406531528782 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (90 to 91)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535118362, 6406535118363

MAD1_183 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 133 Question Id : 6406531533251 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535130953. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535130954. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535130955. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535130956. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535130957. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535130953, 6406535130955, 6406535130956

MAD1_184 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 134 Question Id : 6406531533259 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535130986. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535130987. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535130988. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535130989. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653246674
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535130986, 6406535130988

MAD1_187 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 137 Question Id : 6406531533257 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535130978. ✔ http is the protocol.

406535130979. ✔ 5000 is the port number.

406535130980. ✔ The trailing / represents the base URL .

406535130981. ✖ 5000 is the status code.

Official Answer Key: 6406535130978, 6406535130979, 6406535130980

MAD1_188 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 138 Question Id : 6406531533258 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535130982. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535130983. ✔ If the image file is missing, browsers display it as a broken image.

406535130984. ✖ The ***alt*** attribute is mandatory in HTML5.

406535130985. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653246676
uestion Shuffling Allowed :	No

uestion Id : 6406531533260 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (139 to 140)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535130982, 6406535130983

MAD1_195 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 133 Question Id : 6406531533576 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535131849. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535131850. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535131851. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535131852. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535131853. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535131849, 6406535131851, 6406535131852

MAD1_196 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 134 Question Id : 6406531533584 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535131882. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535131883. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535131884. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535131885. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653246770
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535131882, 6406535131884

MAD1_199 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 137 Question Id : 6406531533582 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535131874. ✔ http is the protocol.

406535131875. ✔ 5000 is the port number.

406535131876. ✔ The trailing / represents the base URL .

406535131877. ✖ 5000 is the status code.

Official Answer Key: 6406535131874, 6406535131875, 6406535131876

MAD1_200 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 138 Question Id : 6406531533583 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

img src="logo.png" alt="Company Logo" />

Which of the following is **true**?

ptions :

406535131878. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535131879. ✔ If the image file is missing, browsers display it as a broken image.

406535131880. ✖ The ***alt*** attribute is mandatory in HTML5.

406535131881. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653246772
uestion Shuffling Allowed :	No

uestion Id : 6406531533585 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (139 to 140)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535131878, 6406535131879

MAD1_206 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 133 Question Id : 6406531533901 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535132745. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535132746. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535132747. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535132748. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535132749. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535132745, 6406535132747, 6406535132748

MAD1_207 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 134 Question Id : 6406531533909 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535132778. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535132779. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535132780. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535132781. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653246866
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535132778, 6406535132780

MAD1_210 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 137 Question Id : 6406531533907 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535132770. ✔ http is the protocol.

406535132771. ✔ 5000 is the port number.

406535132772. ✔ The trailing / represents the base URL .

406535132773. ✖ 5000 is the status code.

Official Answer Key: 6406535132770, 6406535132771, 6406535132772

MAD1_211 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 138 Question Id : 6406531533908 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535132774. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535132775. ✔ If the image file is missing, browsers display it as a broken image.

406535132776. ✖ The ***alt*** attribute is mandatory in HTML5.

406535132777. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653246868
uestion Shuffling Allowed :	No

uestion Id : 6406531533910 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (139 to 140)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535132774, 6406535132775

MAD1_217 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 70 Question Id : 6406531531442 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535125826. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535125827. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535125828. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535125829. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535125830. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535125826, 6406535125828, 6406535125829

MAD1_218 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 71 Question Id : 6406531531450 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535125859. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535125860. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535125861. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535125862. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653246065
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535125859, 6406535125861

MAD1_221 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 74 Question Id : 6406531531448 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535125851. ✔ http is the protocol.

406535125852. ✔ 5000 is the port number.

406535125853. ✔ The trailing / represents the base URL .

406535125854. ✖ 5000 is the status code.

Official Answer Key: 6406535125851, 6406535125852, 6406535125853

MAD1_222 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 75 Question Id : 6406531531449 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535125855. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535125856. ✔ If the image file is missing, browsers display it as a broken image.

406535125857. ✖ The ***alt*** attribute is mandatory in HTML5.

406535125858. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653246067
uestion Shuffling Allowed :	No

uestion Id : 6406531531451 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (76 to 77)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535125855, 6406535125856

MAD1_229 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 70 Question Id : 6406531532325 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535128318. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535128319. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535128320. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535128321. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535128322. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535128318, 6406535128320, 6406535128321

MAD1_230 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 71 Question Id : 6406531532333 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535128351. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535128352. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535128353. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535128354. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653246358
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535128351, 6406535128353

MAD1_233 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 74 Question Id : 6406531532331 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535128343. ✔ http is the protocol.

406535128344. ✔ 5000 is the port number.

406535128345. ✔ The trailing / represents the base URL .

406535128346. ✖ 5000 is the status code.

Official Answer Key: 6406535128343, 6406535128344, 6406535128345

MAD1_234 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 75 Question Id : 6406531532332 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535128347. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535128348. ✔ If the image file is missing, browsers display it as a broken image.

406535128349. ✖ The ***alt*** attribute is mandatory in HTML5.

406535128350. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653246360
uestion Shuffling Allowed :	No

uestion Id : 6406531532334 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (76 to 77)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535128347, 6406535128348

MAD1_240 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 70 Question Id : 6406531532888 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the following Python code snippet.

```
from jinja2 import Template as JTemplate
from string import Template as STemplate

data = {'name1': 'Alice', 'name2': 'name1',
        'age1': '25', 'age2': 'age1',
        'job1': 'Engineer', 'job2': 'job1'}

jt = JTemplate("{{name2}} is {{age2}} years old and works as a {{job2}}.")
result1 = jt.render(data)
print(result1)  #first print statement

st = STemplate(result1)
result2 = st.substitute(data)
print(result2) #second print statement
```

f the above Python code runs on the terminal, which of the following statements is/are correct?

ptions :

406535129888. ✔ The first print statement will output: "name1 is age1 years old and works as a job1."

406535129889. ✖ The second print statement will output: "Alice is 25 years old and works as a Engineer."

406535129890. ✔ Jinja2 templates use double curly braces {{}} for variable substitution

406535129891. ✔ String templates use dollar signs $ for variable substitution in the substitute() method

406535129892. ✖ The code will raise a KeyError exception during execution

Official Answer Key: 6406535129888, 6406535129890, 6406535129891

MAD1_241 | Type: MSQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 71 Question Id : 6406531532896 Question Type : MSQ
orrect Marks : 3 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Which of the following Flask view functions will NOT throw any error for the URL:
`http://127.0.0.1:5000/user?username=mad1&role=admin`?

ptions :

406535129921. ✔

```
@app.route('/user')
def profile():
    creds = request.args
    info = {'username': creds["username"], 'role': creds["role"]}
    return info
```

406535129922. ✖

```
@app.route('/user/<username>/<role>')
def profile(username, role):
    info = {'username': username, 'role': role}
    return info
```

406535129923. ✔

```
@app.route('/user')
def profile():
    return {'message': 'data retrieved successfully'}
```

406535129924. ✖ All of these

ub-Section Number :	5
ub-Section Id :	640653246553
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535129921, 6406535129923

MAD1_244 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 74 Question Id : 6406531532894 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the web URL.
http://localhost:5000/

Which of the following is/are **True**?

ptions :

406535129913. ✔ http is the protocol.

406535129914. ✔ 5000 is the port number.

406535129915. ✔ The trailing / represents the base URL .

406535129916. ✖ 5000 is the status code.

Official Answer Key: 6406535129913, 6406535129914, 6406535129915

MAD1_245 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 75 Question Id : 6406531532895 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question
Consider the following HTML snippet:

```
<img src="logo.png" alt="Company Logo" />
```

Which of the following is **true**?

ptions :

406535129917. ✔ If ***logo.png*** fails to load, the text ***"Company Logo"*** will be shown.

406535129918. ✔ If the image file is missing, browsers display it as a broken image.

406535129919. ✖ The ***alt*** attribute is mandatory in HTML5.

406535129920. ✖ ***Logo.png*** fails to load because the ***<img>*** tag only allows "jpeg/jpg" files.

ub-Section Number :	7
ub-Section Id :	640653246555
uestion Shuffling Allowed :	No

uestion Id : 6406531532897 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

uestion Numbers : (76 to 77)
uestion Label : Comprehension
Consider the following Python code snippet.
Filename: server.py

```python
from flask import Flask, request
import sys

app = Flask(__name__)

@app.route('/api/data', methods=sys.argv[1:])
def handle_data():
    if request.method == 'GET':
        return "<h1>Data retrieved successfully</h1>"
    elif request.method == 'PUT':
        return "<h1>Data updated successfully</h1>"
    elif request.method == 'DELETE':
        return "<h1>Data deleted successfully</h1>"
    else:
        return "<h1>Unsupported operation</h1>"

app.run(debug=True)
```

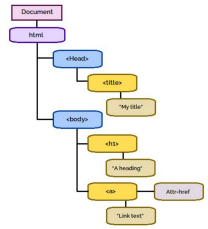
ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535129917, 6406535129918

MAD1_249 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 105 Question Id : 64065368548 Question Type : MCQ Is Question
andatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider the following DOM structure and select the HTML document that correctly represents
e DOM structure .



ptions :

406532239831.
```
<!DOCTYPE html>
<head>
    <title>My title</title>
<body>
    <h1>A heading</h1>
    <a href="link">Link text</a>
```

406532239832.
```
<!DOCTYPE html>
<html>
<head>
    <title>My title</title>
</head>
<body>
    <h1>A heading</h1>
    <a>Link text</a>
</body>
</html>
```

406532239833.
```
<!DOCTYPE html>
<html>
<head>
    <title>My title</title>
<body>
    <h1>A heading</h1>
    <a>Link text</a>
</html>
```

406532239834.
```
<!DOCTYPE html>
<html>
<head>
    <title>My title</title>
</head>
<body>
    <h1>A heading</h1>
    <a href="link">Link text</a>
</body>
</html>
```

Official Answer Key: 6406532239834

MAD1_262 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

Official Answer Key: 6406533360888

MAD1_272 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 113 Question Id : 640653995545 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the flask code below.

```
from flask import Flask

app = Flask(__name__)

@app.route("/employee/<id>")
def get_employee(id):
   return "Employee id : " + str(id)

if __name__ == "__main__":
   app.run(debug=True)
```

Assume that the above flask application is running on http://127.0.0.1:5000 . Which of the following URLs will render **"Not Found"** error?

ptions :

406533360933. ✖ http://127.0.0.1:5000/employee/102

406533360934. ✖ http://127.0.0.1:5000/employee/id=102

406533360935. ✔ http://127.0.0.1:5000/id=102

406533360936. ✔ http://127.0.0.1:5000/102

ub-Section Number :	6
ub-Section Id :	640653146814
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533360935, 6406533360936

MAD1_273 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 114 Question Id : 640653995547 Question Type : MSQ
orrect Marks : 4.5 Max. Selectable Options : 0
uestion Label : Multiple Select Question
onsider the following code snippet.

```
@app.route('/name/<type>')
def prod_detail(type):
    # CODE BLOCK HERE
```

ssume the database has a "product" table represented by class "Product" which has a EXT column "type". If we want the server to return a 404 status code when a user goes to e route '/name/<type>' with a "type" that does not exist in the database, which of the llowing lines would give us the desired output?

ptions :

406533360941. ✖

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html")
return render_template('details.html', type=type)
```

406533360942. ✔

```
type = Product.query.filter_by(type=type).first()
if type is None:
    abort(404)
return render_template('details.html', type=type)
```

406533360943. ✔

```
type = Product.query.filter_by(type=type).first_or_404()
return render_template('details.html', type=type)
```

406533360944. ✔

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html"), 404
return render_template('details.html', type=type)
```

ub-Section Number : 7

Official Answer Key: 6406533360942, 6406533360943, 6406533360944

MAD1_274 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 115 Question Id : 640653995536 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
uppose the value of **Method** in form.html is GET
form method is set to GET) and in the input box
ve write “Application”, after submitting the form
how the URL will look? Remember server is
running on the server, http://127.0.0.1:5000.

ptions :

406533360896. ✔ http://127.0.0.1:5000/?name=Application

406533360897. ✖ http://127.0.0.1:5000/name=Application

406533360898. ✖ http://127.0.0.1:5000/?name

Official Answer Key: 6406533360896

MAD1_275 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 116 Question Id : 640653995537 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the nput box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

ptions :

406533360900. ✖ http://127.0.0.1:5000/?name:Application

406533360901. ✖ http://127.0.0.1:5000/?Application

406533360902. ✖ http://127.0.0.1:5000/?name

406533360903. ✔ http://127.0.0.1:5000/

Java

Official Answer Key: 6406533360903

MAD1_277 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 103 Question Id : 640653995815 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

:onsider the following HTML code and select the correct rendered output from the ollowing.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
</head>
<body>
    <table border="2">
        <tr>
          <th>One</th>
          <td>Four</td>
        </tr>
        <tr>
          <th> Two</th>
          <td> Three</td>
        </tr>
    </table>
</body>
</html>
```

ptions :

406533361672. ✔

One	Four
Two	Three

406533361673. ✖

One	Two
Four	Three

406533361674. ✖

One	Two
Three	Four

406533361675. ✖

One	Four
Two	Three

Official Answer Key: 6406533361672

MAD1_285 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 113 Question Id : 640653995827 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the flask code below.

```
from flask import Flask

app = Flask(__name__)

@app.route("/employee/<id>")
def get_employee(id):
   return "Employee id : " + str(id)

if __name__ == "__main__":
   app.run(debug=True)
```

Assume that the above flask application is running on http://127.0.0.1:5000 . Which of the following URLs will render **"Not Found"** error?

ptions :

406533361717. ✖ http://127.0.0.1:5000/employee/102

406533361718. ✖ http://127.0.0.1:5000/employee/id=102

406533361719. ✔ http://127.0.0.1:5000/id=102

406533361720. ✔ http://127.0.0.1:5000/102

ub-Section Number :	6
ub-Section Id :	640653146903
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533361719, 6406533361720

MAD1_286 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 114 Question Id : 640653995829 Question Type : MSQ
orrect Marks : 4.5 Max. Selectable Options : 0
uestion Label : Multiple Select Question
onsider the following code snippet.

```
@app.route('/name/<type>')
def prod_detail(type):
    # CODE BLOCK HERE
```

ssume the database has a "product" table represented by class "Product" which has a EXT column "type". If we want the server to return a 404 status code when a user goes to e route '/name/<type>' with a "type" that does not exist in the database, which of the llowing lines would give us the desired output?

ptions :

406533361725. ✖

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html")
return render_template('details.html', type=type)
```

406533361726. ✔

```
type = Product.query.filter_by(type=type).first()
if type is None:
    abort(404)
return render_template('details.html', type=type)
```

406533361727. ✔

```
type = Product.query.filter_by(type=type).first_or_404()
return render_template('details.html', type=type)
```

406533361728. ✔

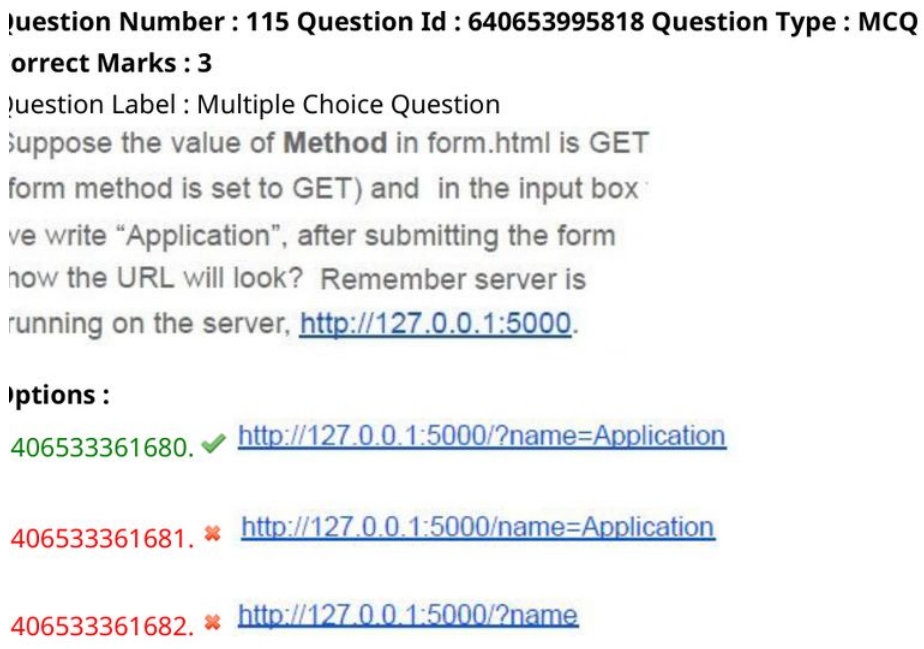
```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html"), 404
return render_template('details.html', type=type)
```

ub-Section Number : 7

Official Answer Key: 6406533361726, 6406533361727, 6406533361728

MAD1_287 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf



uestion Number : 115 Question Id : 640653995818 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
uppose the value of **Method** in form.html is GET
form method is set to GET) and in the input box
ve write "Application", after submitting the form
how the URL will look? Remember server is
running on the server, http://127.0.0.1:5000.

ptions :

406533361680. ✔ http://127.0.0.1:5000/?name=Application

406533361681. ✖ http://127.0.0.1:5000/name=Application

406533361682. ✖ http://127.0.0.1:5000/?name

Official Answer Key: 6406533361680

MAD1_288 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 116 Question Id : 640653995819 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the nput box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

ptions :

406533361684. ✖ http://127.0.0.1:5000/?name:Application

406533361685. ✖ http://127.0.0.1:5000/?Application

406533361686. ✖ http://127.0.0.1:5000/?name

406533361687. ✔ http://127.0.0.1:5000/

Java

Official Answer Key: 6406533361687

MAD1_290 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 103 Question Id : 640653996097 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the following HTML code and select the correct rendered output from the ollowing.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
</head>
<body>
    <table border="2">
        <tr>
          <th>One</th>
          <td>Four</td>
        </tr>
        <tr>
          <th> Two</th>
          <td> Three</td>
        </tr>
    </table>
</body>
</html>
```

ptions :

406533362456. ✔

One	Four
Two	Three

406533362457. ✖

One	Two
Four	Three

406533362458. ✖

One	Two
Three	Four

406533362459. ✖

One	Four
Two	Three

Official Answer Key: 6406533362456

MAD1_298 | Type: MSQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 113 Question Id : 640653996109 Question Type : MSQ
orrect Marks : 2 Max. Selectable Options : 0
uestion Label : Multiple Select Question

Consider the flask code below.

```
from flask import Flask

app = Flask(__name__)

@app.route("/employee/<id>")
def get_employee(id):
   return "Employee id : " + str(id)

if __name__ == "__main__":
   app.run(debug=True)
```

Assume that the above flask application is running on http://127.0.0.1:5000 . Which of the following URLs will render **"Not Found"** error?

ptions :

406533362501. ✗ http://127.0.0.1:5000/employee/102

406533362502. ✗ http://127.0.0.1:5000/employee/id=102

406533362503. ✓ http://127.0.0.1:5000/id=102

406533362504. ✓ http://127.0.0.1:5000/102

ub-Section Number :	6
ub-Section Id :	640653146992
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533362503, 6406533362504

MAD1_299 | Type: MSQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 114 Question Id : 640653996111 Question Type : MSQ
orrect Marks : 4.5 Max. Selectable Options : 0
uestion Label : Multiple Select Question
onsider the following code snippet.

```
@app.route('/name/<type>')
def prod_detail(type):
    # CODE BLOCK HERE
```

ssume the database has a "product" table represented by class "Product" which has a EXT column "type". If we want the server to return a 404 status code when a user goes to e route '/name/<type>' with a "type" that does not exist in the database, which of the llowing lines would give us the desired output?

ptions :

406533362509. ✖

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html")
return render_template('details.html', type=type)
```

406533362510. ✔

```
type = Product.query.filter_by(type=type).first()
if type is None:
    abort(404)
return render_template('details.html', type=type)
```

406533362511. ✔

```
type = Product.query.filter_by(type=type).first_or_404()
return render_template('details.html', type=type)
```

406533362512. ✔

```
type = Product.query.filter_by(type=type).first()
if type is None:
    return render_template("404.html"), 404
return render_template('details.html', type=type)
```

ub-Section Number : 7

Official Answer Key: 6406533362510, 6406533362511, 6406533362512

MAD1_300 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 115 Question Id : 640653996100 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
uppose the value of **Method** in form.html is GET
form method is set to GET) and in the input box
ve write “Application”, after submitting the form
how the URL will look? Remember server is
running on the server, http://127.0.0.1:5000.

ptions :

406533362464. ✔ http://127.0.0.1:5000/?name=Application

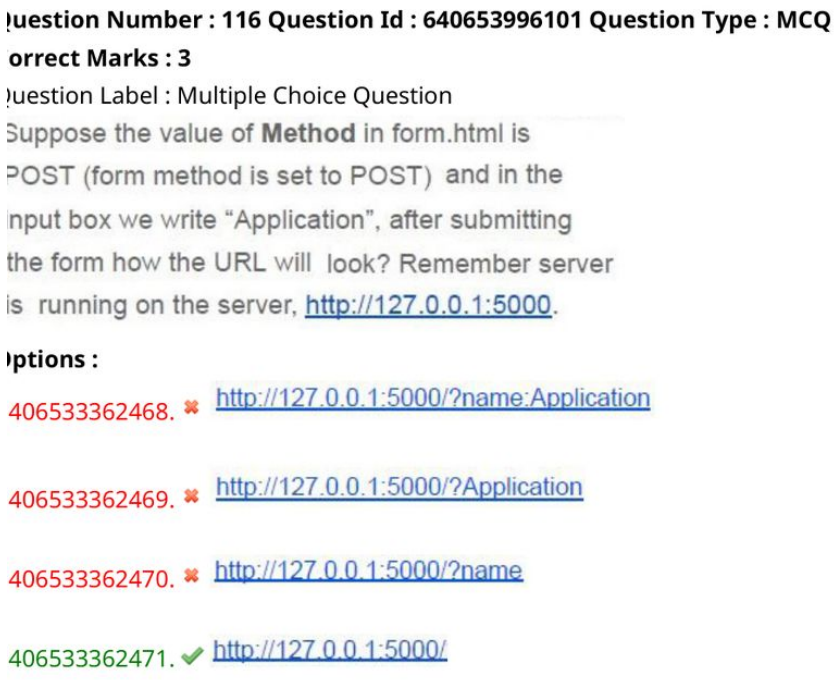
406533362465. ✖ http://127.0.0.1:5000/name=Application

406533362466. ✖ http://127.0.0.1:5000/?name

Official Answer Key: 6406533362464

MAD1_301 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf



uestion Number : 116 Question Id : 640653996101 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Suppose the value of **Method** in form.html is POST (form method is set to POST) and in the nput box we write "Application", after submitting the form how the URL will look? Remember server is running on the server, http://127.0.0.1:5000.

ptions :

406533362468. ✖ http://127.0.0.1:5000/?name:Application

406533362469. ✖ http://127.0.0.1:5000/?Application

406533362470. ✖ http://127.0.0.1:5000/?name

406533362471. ✔ http://127.0.0.1:5000/

Java

Official Answer Key: 6406533362471

• 2.3 CSS Selectors, Specificity & Cascade Rules (5 Questions)

MAD1_036 | Type: MCQ | Marks: +4.5

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 116 Question Id : 640653470037 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5

uestion Label : Multiple Choice Question

ead the statements given below carefully and select the correct option.

tatement 1: If an element that belongs to two different classes is styled externally using both tl
lasses, then for the same attribute, it will acquire styling from the latest class in order.
tatement 2: If an element having an ID and a class is styled externally using both its ID and the
lass, then for the same attribute, it will acquire styling from the latest selector in order.

ptions :

406531562002. ✖ Both statements 1 and 2 are correct

406531562003. ✖ Both statements 1 and 2 are incorrect

406531562004. ✔ Statement 1 is correct but statement 2 is incorrect

406531562005. ✖ Statement 2 is correct but statement 1 is incorrect

Official Answer Key: 6406531562004

MAD1_048 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 69 Question Id : 6406531113761 Question Type : MCQ

orrect Marks : 2

uestion Label : Multiple Choice Question

ead the statements given below carefully and select the correct option.

tatement 1: If an element that belongs to two different classes is styled externally using both tl
lasses, then for the same attribute, it will acquire styling from the latest class in order.
tatement 2: If an element having an ID and a class is styled externally using both its ID and the
lass, then for the same attribute, it will acquire styling from the ID selector.

ptions :

406533774148. ✔ Both statements 1 and 2 are correct

406533774149. ✖ Both statements 1 and 2 are incorrect

406533774150. ✖ Statement 1 is correct but statement 2 is incorrect

406533774151. ✖ Statement 2 is correct but statement 1 is incorrect

ub-Section Number :	3
ub-Section Id :	640653168509
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533774148

MAD1_063 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 69 Question Id : 6406531114388 Question Type : MCQ

orrect Marks : 2

uestion Label : Multiple Choice Question

ead the statements given below carefully and select the correct option.

tatement 1: If an element that belongs to two different classes is styled externally using both tl
lasses, then for the same attribute, it will acquire styling from the latest class in order.
tatement 2: If an element having an ID and a class is styled externally using both its ID and the
lass, then for the same attribute, it will acquire styling from the ID selector.

ptions :

406533775961. ✔ Both statements 1 and 2 are correct

406533775962. ✖ Both statements 1 and 2 are incorrect

406533775963. ✖ Statement 1 is correct but statement 2 is incorrect

406533775964. ✖ Statement 2 is correct but statement 1 is incorrect

ub-Section Number :	3
ub-Section Id :	640653168713
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533775961

MAD1_077 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 69 Question Id : 6406531114101 Question Type : MCQ

orrect Marks : 2

uestion Label : Multiple Choice Question

ead the statements given below carefully and select the correct option.

tatement 1: If an element that belongs to two different classes is styled externally using both tl
lasses, then for the same attribute, it will acquire styling from the latest class in order.
tatement 2: If an element having an ID and a class is styled externally using both its ID and the
lass, then for the same attribute, it will acquire styling from the ID selector.

ptions :

406533775132. ✔ Both statements 1 and 2 are correct

406533775133. ✖ Both statements 1 and 2 are incorrect

406533775134. ✖ Statement 1 is correct but statement 2 is incorrect

406533775135. ✖ Statement 2 is correct but statement 1 is incorrect

ub-Section Number :	3
ub-Section Id :	640653168614
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533775132

MAD1_131 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 41 Question Id : 640653825026 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider the following HTML document.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Span</title>
    <style>
        span{
            color: blue;
            Background-color: lime;
        }
    </style>
</head>
<body>
    <span id="my_span">MAD-I is a Diploma level course.</span>
    <span>SE is a Degree level course.</span>
</body>
</html>
```

/hat will be the rendered output if an additional style is added via the ID selector given below?

```
#my_span{
        width: 1200px;
        Background-color: lightpink;
       }
```

ptions :

406532774751. ✖ MAD-I is a Diploma level course. SE is a Degree level course.

406532774752. ✔ MAD-I is a Diploma level course. SE is a Degree level course.

406532774753. ✖ MAD-I is a Diploma level course. SE is a Degree level course.

406532774754. ✖ MAD-I is a Diploma level course.
SE is a Degree level course.

Official Answer Key: 6406532774752

• 2.4 CSS Box Model & Layouts (Flexbox, Positioning & Spacing) (16 Questions)

MAD1_025 | Type: MCQ | Marks: +4.5

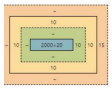
Source Exam(s): IIT M DIPLOMA QUIZ1 EXAM QPF1.pdf

uestion Number : 118 Question Id : 640653345252 Question Type : MCQ Is Question
andatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5
uestion Label : Multiple Choice Question
onsider the following HTML document with an embedded style sheet.

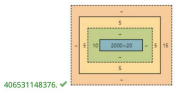
```
<!DOCTYPE html>
<html>
    <head>
        <title>Quiz 1</title>
        <style type="text/css">
            div{
                padding-left: 10px;
                margin-right: 15px;
                border-style: solid;
                border-width: 5px;
                width: 2000px;
                height: 20px;
                }
        </style>
    </head>
    <body>
        <div>My first Div element</div>
    </body>
</html>
```

/hich of the following figures correctly represents the box model of the above HTML document?
ptions :

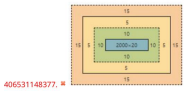
406531148375.



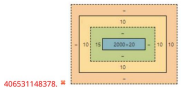
406531148376. ✔



406531148377.



406531148378.



Official Answer Key: 6406531148376

MAD1_037 | Type: MCQ | Marks: +4.5

Source Exam(s): Quiz 1 Jan23 .pdf, Quiz1 May23 .pdf

uestion Number : 117 Question Id : 640653470039 Question Type : MCQ Is Question
andatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 4.5

uestion Label : Multiple Choice Question

onsider the following HTML document.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        #mybox{
            border-style: dashed;
            background-color: orange;
        }
        .boxes{
            border-style: solid;
            border-color: blue;
        }
        div{
            width: 200px;
            border: 4px dotted yellow;
        }
    </style>
</head>
<body>
    <div id="mybox" class="boxes" style="background-color: limegreen;">
        <h1>Hello MAD-I</h1>
    </div>
</body>
</html>
```

ow will the browser render the above HTML document?

ptions :

406531562010. ✗

Hello MAD-I

406531562011. ✗ Hello MAD-I

406531562012. ✓ Hello MAD-I

406531562013. ✗ Hello MAD-I

ub-Section Number :	5
ub-Section Id :	64065367683
uestion Shuffling Allowed :	Yes
Section Default? :	null

Official Answer Key: 6406531562012

MAD1_049 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 71 Question Id : 6406531113763 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the following HTML document:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        #container {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
```

Official Answer Key: 6406533774156

MAD1_064 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 71 Question Id : 6406531114390 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the following HTML document:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        #container {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
```

Official Answer Key: 6406533775969

MAD1_078 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 71 Question Id : 6406531114103 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the following HTML document:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        #container {
            border-style: dashed;
            background-color: red;
        }
        .content {
            border-style: solid;
            border-color: green;
            background-color: orange;
        }
        div {
            width: 250px;
            border: 3px dotted black;
            border-color: yellow;
        }
    </style>
</head>
<body>
```

Official Answer Key: 6406533775140

MAD1_090 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz1 May23 .pdf

uestion Number : 105 Question Id : 640653577864 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

onsider the image given below.

Forgot email?

When a user enters an invalid email or phone number, an error message is displayed which clearly tells him/her about what went wrong. Which usability heuristic principle best describes the above behavior?

ptions :

406531929691. ✖ Aesthetic and minimalist design

406531929692. ✖ User control and freedom

406531929693. ✔ User diagnosis and recovery from errors

406531929694. ✖ Flexibility and ease of use

Official Answer Key: 6406531929693

MAD1_149 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 86 Question Id : 6406531527812 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535115576. ✗ 200 px
406535115577. ✗ 230 px
406535115578. ✓ 240 px
406535115579. ✗ 250 px

Official Answer Key: 6406535115578

MAD1_162 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 86 Question Id : 6406531528115 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535116419. ✗ 200 px
406535116420. ✗ 230 px
406535116421. ✓ 240 px
406535116422. ✗ 250 px

Official Answer Key: 6406535116421

MAD1_173 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 86 Question Id : 6406531528777 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535118350. ✗ 200 px
406535118351. ✗ 230 px
406535118352. ✓ 240 px
406535118353. ✗ 250 px

Official Answer Key: 6406535118352

MAD1_185 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 135 Question Id : 6406531533255 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535130970. ✗ 200 px
406535130971. ✗ 230 px
406535130972. ✓ 240 px
406535130973. ✗ 250 px

Official Answer Key: 6406535130972

MAD1_197 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 135 Question Id : 6406531533580 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535131866. ✗ 200 px
406535131867. ✗ 230 px
406535131868. ✓ 240 px
406535131869. ✗ 250 px

Official Answer Key: 6406535131868

MAD1_208 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 135 Question Id : 6406531533905 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535132762. ✗ 200 px
406535132763. ✗ 230 px
406535132764. ✓ 240 px
406535132765. ✗ 250 px

Official Answer Key: 6406535132764

MAD1_219 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 72 Question Id : 6406531531446 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535125843. ✗ 200 px
406535125844. ✗ 230 px
406535125845. ✓ 240 px
406535125846. ✗ 250 px

Official Answer Key: 6406535125845

MAD1_231 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 72 Question Id : 6406531532329 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :

406535128335. ✗ 200 px
406535128336. ✗ 230 px
406535128337. ✓ 240 px
406535128338. ✗ 250 px

Official Answer Key: 6406535128337

MAD1_242 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 72 Question Id : 6406531532892 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following HTML and CSS code:

```
<html>
  <head>
    <style>
      div {
        width: 0px;
        margin: 0px;
      }
      .container {
        width: 200px;
        padding: 15px;
        border: 5px solid black;
      }
      #box {
        margin: 10px;
      }
    </style>
  </head>
  <body>
    <div class="container" id="box">Hello World</div>
  </body>
</html>
```

What is the **total width (approximately)** of the `div` with `class="container"` and `id="box"`, including **content, padding, and border** but **excluding margin**?

ptions :
406535129905. ✗ 200 px
406535129906. ✗ 230 px
406535129907. ✓ 240 px
406535129908. ✗ 250 px

Official Answer Key: 6406535129907

MAD1_247 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 102 Question Id : 640653668542 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

hat does the term “responsive” mean in the context of web design?

ptions :

406532239807. ✖ The ability of a webpage to give visual feedback on any kind of user input.

406532239808. ✖ The ability of a webpage to change its color scheme to “night mode”.

406532239809. ✔ The ability of a webpage to alter its layout and appearance to suit different creen widths, resolutions, etc.

406532239810. ✖ All of these.

Official Answer Key: 6406532239809

WEEK 3: BACKEND PYTHON, FLASK ESSENTIALS, ROUTING & REQUEST HANDLING

• 3.1 Python Web Fundamentals & Command-Line Arguments (29 Questions)

MAD1_009 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 72 Question Id : 6406531298441 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following Python code snippet:

main.py

```
import sys
def process(v):
    # Missing code 1
v = sys.argv
# Missing code 2
print(output)
```

We are executing above main.py in command line with command `python main.py 1 2 3 4 5` .
Which of the following codes when replaced at #Missing code 1 and #Missing code 2 will render the output below.

1,2,3,4,5

ptions :

#Missing code 1
```
st = ",".join(v)
return st
```
#Missing code 2
```
output = process(v)
```
406534366211. ✗

#Missing code 1
```
st = str(join(v),",")
return st
```
#Missing code 2
```
output = process(v)
```
406534366212. ✗

#Missing code 1
```
st = str(join(v),",")
return st
```
#Missing code 2
```
output = process(v[1:])
```
406534366213. ✗

#Missing code 1
```
st = ",".join(v)
return st
```
#Missing code 2
```
output = process(v[1:])
```
406534366214. ✓

ub-Section Number : 5
ub-Section Id : 640653206741
uestion Shuffling Allowed : Yes

Official Answer Key: 6406534366214

MAD1_013 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 76 Question Id : 6406531298446 Question Type : MCQ

orrect Marks : 2

uestion Label : Multiple Choice Question

What will be the output for command line input,

`ython webapp.py react type ?`

ptions :

406534366222. ✔ `react is a client-side javascript framework`

406534366223. ✖ `react is a server-side python framework`

406534366224. ✖ `Invalid second argument or language not found!`

406534366225. ✖ `Invalid command`

Official Answer Key: 6406534366222

MAD1_014 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 77 Question Id : 6406531298447 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

What will be the output for command line input, `python webapp.py django python` ?

ptions :

406534366226. ✖ `django is a client-side javascript framework`

406534366227. ✖ `django is a server-side python framework`

406534366228. ✔ `Invalid second argument or language not found!`

406534366229. ✖ `Invalid command`

ub-Section Number :	7
ub-Section Id :	640653206743
uestion Shuffling Allowed :	No

Official Answer Key: 6406534366228

MAD1_034 | Type: MCQ | Marks: +2.0

Source Exam(s): Quiz 1 Jan23 .pdf

uestion Number : 113 Question Id : 640653470034 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 2

uestion Label : Multiple Choice Question

Find the output of the python program for the given command line arguments

Program:

```
import sys

a,b,c = sys.argv[1], sys.argv[2], sys.argv[3]
print(b,c)
```

Execution with command line arguments:
$ python3 cl.py 10 11 12

ptions :

406531561990. ✖ cl.py 10

406531561991. ✖ 10 11

406531561992. ✔ 11 12

406531561993. ✖ cl.py 11

Official Answer Key: 6406531561992

MAD1_052 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 74 Question Id : 6406531113766 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the following Python code snippet:

ython code: "main.py"

```
import sys
movie_ratings = {
   "Kalki": [5, 1, 2, 1, 1],
   "Jailer": [7, 3, 1, 2, 1],
   "Leo": [4, 3, 2, 2, 1],
}

def do_something(movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return "Movie not found"

args = sys.argv
print(do_something(args[1]))
```

he above file will result in the output as **"2.8"**, for which of the following command line input?

ptions :

406533774168. ✖ python main.py Kalki
406533774169. ✖ python main.py Leo
406533774170. ✖ python main.py Pushpa2
406533774171. ✔ python main.py Jailer

Official Answer Key: 6406533774171

MAD1_053 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 75 Question Id : 6406531113767 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider two python files, mad1.py and mad2.py with following code snippets.
File1: mad1.py

```
import sys
import mad2
print('Welcome' + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print('Welcome' + ' ' + sys.argv[0])
```

What is the output of the following command "python mad1.py mad2.py MAD-I MAD-II"?

ptions :

406533774172. ✖
Welcome MAD-I.py
Welcome MAD-II.py

406533774173. ✖
Welcome mad2.py
Welcome mad1.py

406533774174. ✔
Welcome mad1.py
Welcome mad2.py

406533774175. ✖
Welcome MAD-II.py
Welcome MAD-I.py

ub-Section Number :	5
ub-Section Id :	640653168511
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533774174

MAD1_066 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 74 Question Id : 6406531114393 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the following Python code snippet:

ython code: "main.py"

```
import sys
movie_ratings = {
   "Kalki": [5, 1, 2, 1, 1],
   "Jailer": [7, 3, 1, 2, 1],
   "Leo": [4, 3, 2, 2, 1],
}

def do_something(movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return "Movie not found"

args = sys.argv
print(do_something(args[1]))
```

he above file will result in the output as **"2.8"**, for which of the following command line input?

ptions :

406533775981. ✖ python main.py Kalki
406533775982. ✖ python main.py Leo
406533775983. ✖ python main.py Pushpa2
406533775984. ✔ python main.py Jailer

Official Answer Key: 6406533775984

MAD1_067 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 75 Question Id : 6406531114394 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider two python files, mad1.py and mad2.py with following code snippets.
File1: mad1.py

```
import sys
import mad2
print('Welcome' + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print('Welcome' + ' ' + sys.argv[0])
```

What is the output of the following command "python mad1.py mad2.py MAD-I MAD-II"?

ptions :

406533775985. ✖
Welcome MAD-I.py
Welcome MAD-II.py

406533775986. ✖
Welcome mad2.py
Welcome mad1.py

406533775987. ✔
Welcome mad1.py
Welcome mad2.py

406533775988. ✖
Welcome MAD-II.py
Welcome MAD-I.py

ub-Section Number :	5
ub-Section Id :	640653168715
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533775987

MAD1_080 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 74 Question Id : 6406531114106 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code snippet:

Python code: "main.py"

```
import sys
movie_ratings = {
   "Kalki": [5, 1, 2, 1, 1],
   "Jailer": [7, 3, 1, 2, 1],
   "Leo": [4, 3, 2, 2, 1],
}

def do_something(movie_name):
   avg_ratings = {}
   for movie in movie_ratings:
       if movie == movie_name:
           avg_ratings[movie] = sum(movie_ratings[movie]) /
len(movie_ratings[movie])
           return avg_ratings
   return "Movie not found"

args = sys.argv
print(do_something(args[1]))
```

The above file will result in the output as **"2.8"**, for which of the following command line input?

ptions :

406533775152. ✖ python main.py Kalki
406533775153. ✖ python main.py Leo
406533775154. ✖ python main.py Pushpa2
406533775155. ✔ python main.py Jailer

Official Answer Key: 6406533775155

MAD1_081 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 75 Question Id : 6406531114107 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider two python files, mad1.py and mad2.py with following code snippets.
File1: mad1.py

```
import sys
import mad2
print('Welcome' + ' ' + sys.argv[1])
```

File2: mad2.py

```
import sys
print('Welcome' + ' ' + sys.argv[0])
```

What is the output of the following command "python mad1.py mad2.py MAD-I MAD-II"?

ptions :

406533775156. ✖ Welcome MAD-I.py
Welcome MAD-II.py

406533775157. ✖ Welcome mad2.py
Welcome mad1.py

406533775158. ✔ Welcome mad1.py
Welcome mad2.py

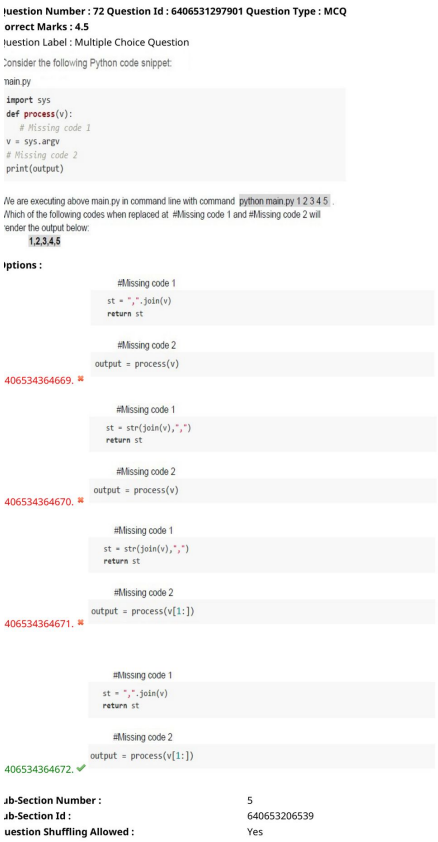
406533775159. ✖ Welcome MAD-II.py
Welcome MAD-I.py

ub-Section Number :	5
ub-Section Id :	640653168616
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533775158

MAD1_102 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf



uestion Number : 72 Question Id : 6406531297901 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following Python code snippet:

nain.py

```
import sys
def process(v):
    # Missing code 1
v = sys.argv
# Missing code 2
print(output)
```

Ve are executing above main.py in command line with command python main.py 1 2 3 4 5 .
Which of the following codes when replaced at #Missing code 1 and #Missing code 2 will render the output below:
1,2,3,4,5

ptions :

#Missing code 1

```
st = ",".join(v)
return st
```

#Missing code 2

```
output = process(v)
```

406534364669. ✗

#Missing code 1

```
st = str(join(v),",")
return st
```

#Missing code 2

```
output = process(v)
```

406534364670. ✗

#Missing code 1

```
st = str(join(v),",")
return st
```

#Missing code 2

```
output = process(v[1:])
```

406534364671. ✗

#Missing code 1

```
st = ",".join(v)
return st
```

#Missing code 2

```
output = process(v[1:])
```

406534364672. ✓

ub-Section Number :	5
ub-Section Id :	640653206539
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406534364672

MAD1_105 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 76 Question Id : 6406531297906 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
What will be the output for command line input,
ython webapp.py react type ?

ptions :

406534364680. ✔ react is a client-side javascript framework

406534364681. ✖ react is a server-side python framework

406534364682. ✖ Invalid second argument or language not found!

406534364683. ✖ Invalid command

Official Answer Key: 6406534364680

MAD1_106 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 77 Question Id : 6406531297907 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

What will be the output for command line input, `python webapp.py django python` ?

ptions :

406534364684. ✖ `django is a client-side javascript framework`

406534364685. ✖ `django is a server-side python framework`

406534364686. ✔ `Invalid second argument or language not found!`

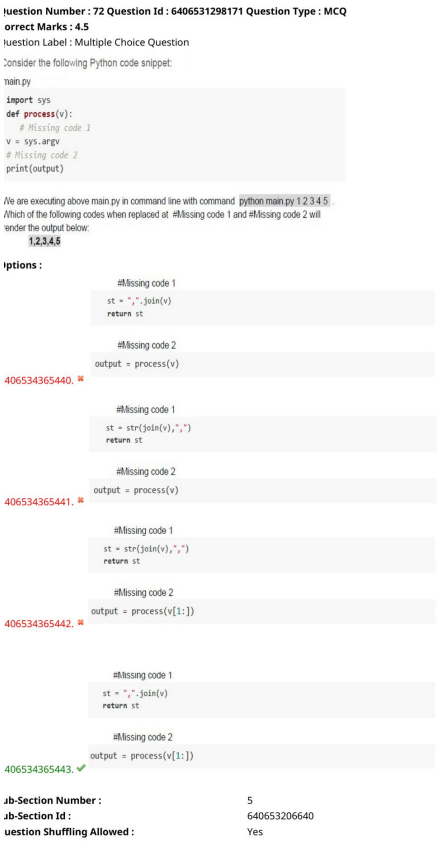
406534364687. ✖ `Invalid command`

ub-Section Number :	7
ub-Section Id :	640653206541
uestion Shuffling Allowed :	No

Official Answer Key: 6406534364686

MAD1_117 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf



uestion Number : 72 Question Id : 6406531298171 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following Python code snippet:

nain.py

```
import sys
def process(v):
    # Missing code 1
v = sys.argv
# Missing code 2
print(output)
```

Ve are executing above main.py in command line with command python main.py 1 2 3 4 5 .
Which of the following codes when replaced at #Missing code 1 and #Missing code 2 will
render the output below:
1,2,3,4,5

ptions :

#Missing code 1
```
st = ",".join(v)
return st
```
#Missing code 2
```
output = process(v)
```
406534365440. ✗

#Missing code 1
```
st = str(join(v),",")
return st
```
#Missing code 2
```
output = process(v)
```
406534365441. ✗

#Missing code 1
```
st = str(join(v),",")
return st
```
#Missing code 2
```
output = process(v[1:])
```
406534365442. ✗

#Missing code 1
```
st = ",".join(v)
return st
```
#Missing code 2
```
output = process(v[1:])
```
406534365443. ✓

ub-Section Number :	5
ub-Section Id :	640653206640
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406534365443

MAD1_120 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 76 Question Id : 6406531298176 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
What will be the output for command line input,
ython webapp.py react type ?

ptions :

406534365451. ✔ react is a client-side javascript framework

406534365452. ✖ react is a server-side python framework

406534365453. ✖ Invalid second argument or language not found!

406534365454. ✖ Invalid command

Official Answer Key: 6406534365451

MAD1_121 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 77 Question Id : 6406531298177 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

What will be the output for command line input, `python webapp.py django python` ?

ptions :

406534365455. ✖ `django is a client-side javascript framework`

406534365456. ✖ `django is a server-side python framework`

406534365457. ✔ `Invalid second argument or language not found!`

406534365458. ✖ `Invalid command`

ub-Section Number :	7
ub-Section Id :	640653206642
uestion Shuffling Allowed :	No

Official Answer Key: 6406534365457

MAD1_132 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf

uestion Number : 42 Question Id : 640653825031 Question Type : MCQ
orrect Marks : 3

uestion Label : Multiple Choice Question
onsider the following Python code snippet.
ilename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return f"Hello from {sys.argv[1]} method"

    elif request.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debug = True)
```

the above flask app is run using the command `python app.py POST GET` in one erminal, what will be the output on another terminal for command; `url -X POST http://127.0.0.1:5000/home` ?

ptions :

406532774771. ✗ `Hello from POST method`

406532774772. ✓ `Hello from GET method`

406532774773. ✗ `Please enter a valid HTTP method`

406532774774. ✗ `Method not Allowed`

Official Answer Key: 6406532774772

MAD1_133 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 43 Question Id : 640653825032 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
onsider the following Python code snippet.

ilename: module.py

```
from pyhtml import *
import sys
var = sys.argv[1:]

def list_func(stack):
    for tech in stack:
        yield span(tech)

doc = html(
        head(title('My Document')),
        body(
            h3('Frontend technologies'),
            list_func(var[::-1])
        )
)
print(doc.render())
```

ow will the browser render the output of the above code if it is run on the terminal sing the command `python module.py HTML CSS JS VUE`?

Official Answer Key: 6406532774776

MAD1_139 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 42 Question Id : 640653827500 Question Type : MCQ
orrect Marks : 3

uestion Label : Multiple Choice Question
onsider the following Python code snippet.
ilename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return f"Hello from {sys.argv[1]} method"

    elif request.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debug = True)
```

the above flask app is run using the command `python app.py POST GET` in one erminal, what will be the output on another terminal for command; `url -X POST http://127.0.0.1:5000/home` ?

ptions :

406532781295. ✖ `Hello from POST method`

406532781296. ✔ `Hello from GET method`

406532781297. ✖ `Please enter a valid HTTP method`

406532781298. ✖ `Method not Allowed`

Official Answer Key: 6406532781296

MAD1_141 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 7 July 2024.pdf

uestion Number : 42 Question Id : 640653827788 Question Type : MCQ
orrect Marks : 3

uestion Label : Multiple Choice Question
onsider the following Python code snippet.
ilename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return f"Hello from {sys.argv[1]} method"

    elif request.method == 'POST':
        return f"Hello from {sys.argv[2]} method"

    else:
        return "Please enter a valid HTTP method"

app.run(debug = True)
```

the above flask app is run using the command `python app.py POST GET` in one erminal, what will be the output on another terminal for command;
`url -X POST http://127.0.0.1:5000/home` ?

ptions :

406532782091. ✖ `Hello from POST method`

406532782092. ✔ `Hello from GET method`

406532782093. ✖ `Please enter a valid HTTP method`

406532782094. ✖ `Method not Allowed`

Official Answer Key: 6406532782092

MAD1_145 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 82 Question Id : 6406531527810 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535115568. ✖
```
Module: utils.py
Processing: data.txt
```

406535115569. ✖
```
Processing: data.txt
Module: utils.py
```

406535115570. ✔
```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535115570

MAD1_158 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 82 Question Id : 6406531528113 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535116411. ✖
```
Module: utils.py
Processing: data.txt
```

406535116412. ✖
```
Processing: data.txt
Module: utils.py
```

406535116413. ✔
```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535116413

MAD1_169 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 82 Question Id : 6406531528775 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535118342. ✖
```
Module: utils.py
Processing: data.txt
```

406535118343. ✖
```
Processing: data.txt
Module: utils.py
```

406535118344. ✔
```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535118344

MAD1_181 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 131 Question Id : 6406531533253 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535130962. ✖
```
Module: utils.py
Processing: data.txt
```

406535130963. ✖
```
Processing: data.txt
Module: utils.py
```

406535130964. ✔
```
Module: main.py
Processing: output.txt
```

406535130965. ✖
```
Processing: output.txt
Module: main.py
```

Official Answer Key: 6406535130964

MAD1_193 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 131 Question Id : 6406531533578 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535131858. ✖
```
Module: utils.py
Processing: data.txt
```

406535131859. ✖
```
Processing: data.txt
Module: utils.py
```

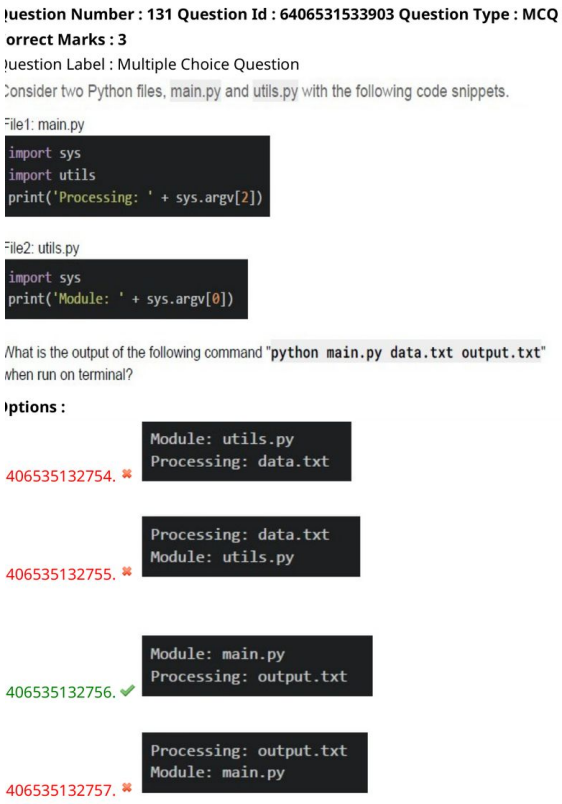
406535131860. ✔
```
Module: main.py
Processing: output.txt
```

406535131861. ✖
```
Processing: output.txt
Module: main.py
```

Official Answer Key: 6406535131860

MAD1_204 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf



uestion Number : 131 Question Id : 6406531533903 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535132754. ✖
```
Module: utils.py
Processing: data.txt
```

406535132755. ✖
```
Processing: data.txt
Module: utils.py
```

406535132756. ✔
```
Module: main.py
Processing: output.txt
```

406535132757. ✖
```
Processing: output.txt
Module: main.py
```

Official Answer Key: 6406535132756

MAD1_215 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 68 Question Id : 6406531531444 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535125835. ✖
```
Module: utils.py
Processing: data.txt
```

406535125836. ✖
```
Processing: data.txt
Module: utils.py
```

406535125837. ✔
```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535125837

MAD1_227 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 68 Question Id : 6406531532327 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535128327. ✖
```
Module: utils.py
Processing: data.txt
```

406535128328. ✖
```
Processing: data.txt
Module: utils.py
```

406535128329. ✔
```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535128329

MAD1_238 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 68 Question Id : 6406531532890 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Consider two Python files, main.py and utils.py with the following code snippets.

File1: main.py

```
import sys
import utils
print('Processing: ' + sys.argv[2])
```

File2: utils.py

```
import sys
print('Module: ' + sys.argv[0])
```

What is the output of the following command "`python main.py data.txt output.txt`" when run on terminal?

ptions :

406535129897. ✖
```
Module: utils.py
Processing: data.txt
```

406535129898. ✖
```
Processing: data.txt
Module: utils.py
```

406535129899. ✔
```
Module: main.py
Processing: output.txt
```

Official Answer Key: 6406535129899

• 3.2 Flask Application Setup & App Configuration (31 Questions)

MAD1_001 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 64 Question Id : 6406531298433 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
What will be the output of the following HTML code?

```
<!DOCTYPE html>
<html>
<head>
    <title>Test List</title>
</head>
<body>
    <dl>
        <dt>Flask</dt>
        <dd>A Python web framework.</dd>
        <dt>SQLAlchemy</dt>
        <dd>A database toolkit for Python.</dd>
    </dl>
</body>
</html>
```

ptions :

406534366179. ✔
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

406534366180. ✖
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

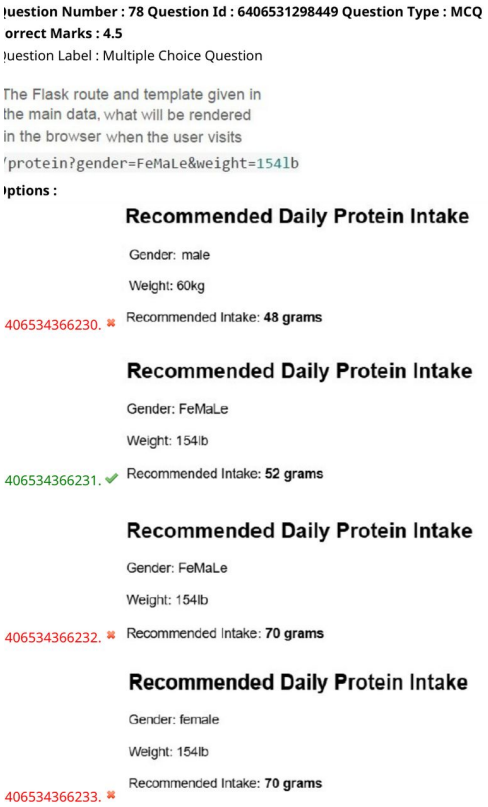
406534366181. ✖
Flask
- A Python web framework.
SQLAlchemy
- A database toolkit for Python.

406534366182. ✖
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

Official Answer Key: 6406534366179

MAD1_015 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf



Official Answer Key: 6406534366231

MAD1_016 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 79 Question Id : 6406531298450 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he Flask route and template given in
he main data, what will be rendered
n the browser when the user visits
'protein?gender=FEMALE&weight=60

ptions :

406534366234. ✖

Recommended Daily Protein Intake

Gender:

Weight:

Recommended Intake: **grams**

406534366235. ✔

Recommended Daily Protein Intake

Gender: FEMALE

Weight: 60

Recommended Intake: **45 grams**

Recommended Daily Protein Intake

Official Answer Key: 6406534366235

MAD1_041 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024.pdf

uestion Number : 133 Question Id : 640653737405 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

latch the URLs in column A with their correct rendered output in Column B with the correct

pp.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

ptions :

406532467942. ✖ 1-b, 2-d, 3-a, 4-c

406532467943. ✖ 1-b, 2-a, 3-c, 4-d

406532467944. ✔ 1-d, 2-b, 3-a, 4- c

406532467945. ✖ 1-d, 2-a, 3-c, 4-b

MLF

Official Answer Key: 6406532467944

MAD1_043 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024.pdf

uestion Number : 133 Question Id : 640653737691 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

latch the URLs in column A with their correct rendered output in Column B with the correct

pp.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

ptions :

406532468766. ✖ 1-b, 2-d, 3-a, 4-c

406532468767. ✖ 1-b, 2-a, 3-c, 4-d

406532468768. ✔ 1-d, 2-b, 3-a, 4- c

406532468769. ✖ 1-d, 2-a, 3-c, 4-b

MLF

Official Answer Key: 6406532468768

MAD1_045 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024.pdf

uestion Number : 133 Question Id : 640653737977 Question Type : MCQ Is Question

landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi

ime : 0

orrect Marks : 3

uestion Label : Multiple Choice Question

latch the URLs in column A with their correct rendered output in Column B with the correct

pp.py code for the flask application given in the previous question.

Column A	Column B
1. http://127.0.0.1:5000/greet?name=MADI&guest=MADII	a. **Hello, Guest!**
2. http://127.0.0.1:5000/greet?name=MADII&guest=MADI	b. **Hello, MADII!**
3. http://127.0.0.1:5000/greet?	c. **Hello, Guest!**
4. http://127.0.0.1:5000/greet?guest=MADI	d. **Hello, MADI!**

ptions :

406532469590. ✖ 1-b, 2-d, 3-a, 4-c

406532469591. ✖ 1-b, 2-a, 3-c, 4-d

406532469592. ✔ 1-d, 2-b, 3-a, 4- c

406532469593. ✖ 1-d, 2-a, 3-c, 4-b

MLF

Official Answer Key: 6406532469592

MAD1_094 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 64 Question Id : 6406531297893 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
What will be the output of the following HTML code?

```
<!DOCTYPE html>
<html>
<head>
    <title>Test List</title>
</head>
<body>
    <dl>
        <dt>Flask</dt>
        <dd>A Python web framework.</dd>
        <dt>SQLAlchemy</dt>
        <dd>A database toolkit for Python.</dd>
    </dl>
</body>
</html>
```

ptions :

406534364637. ✔
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

406534364638. ✖
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

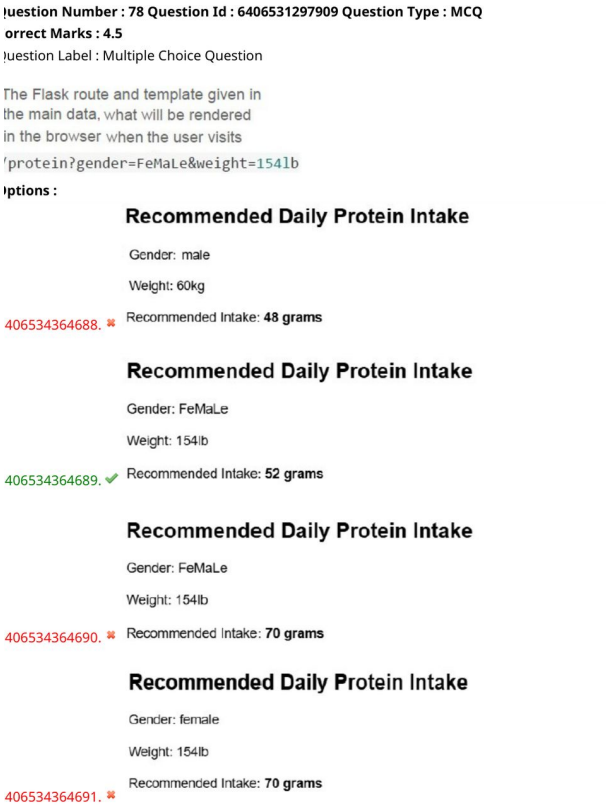
406534364639. ✖
Flask
- A Python web framework.
SQLAlchemy
- A database toolkit for Python.

406534364640. ✖
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

Official Answer Key: 6406534364637

MAD1_107 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf



Official Answer Key: 6406534364689

MAD1_108 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 79 Question Id : 6406531297910 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he Flask route and template given in
he main data, what will be rendered
n the browser when the user visits
`protein?gender=FEMALE&weight=60`

ptions :

406534364692. ✖

Recommended Daily Protein Intake

Gender:

Weight:

Recommended Intake: **grams**

406534364693. ✔

Recommended Daily Protein Intake

Gender: FEMALE

Weight: 60

Recommended Intake: **45 grams**

Recommended Daily Protein Intake

Official Answer Key: 6406534364693

MAD1_109 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 64 Question Id : 6406531298163 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
What will be the output of the following HTML code?

```
<!DOCTYPE html>
<html>
<head>
    <title>Test List</title>
</head>
<body>
    <dl>
        <dt>Flask</dt>
        <dd>A Python web framework.</dd>
        <dt>SQLAlchemy</dt>
        <dd>A database toolkit for Python.</dd>
    </dl>
</body>
</html>
```

ptions :

406534365408. ✔
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

406534365409. ✖
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

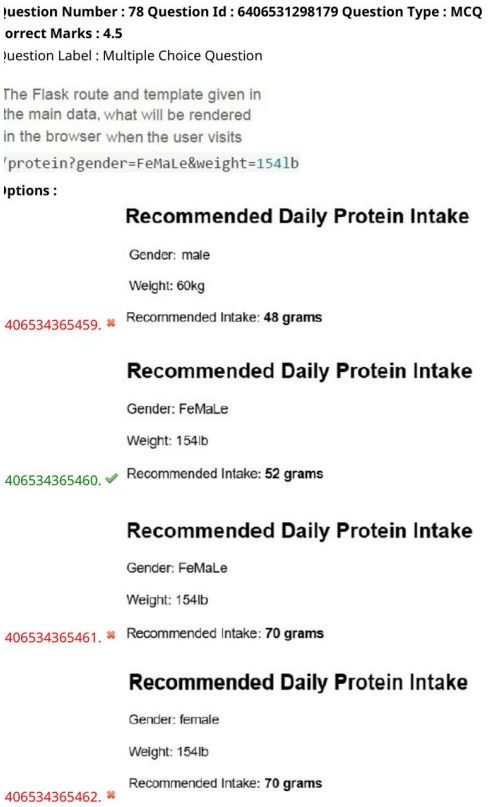
406534365410. ✖
Flask
- A Python web framework.
SQLAlchemy
- A database toolkit for Python.

406534365411. ✖
Flask
A Python web framework.
SQLAlchemy
A database toolkit for Python.

Official Answer Key: 6406534365408

MAD1_122 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf



Official Answer Key: 6406534365460

MAD1_123 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 79 Question Id : 6406531298180 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

he Flask route and template given in
he main data, what will be rendered
n the browser when the user visits
'protein?gender=FEMALE&weight=60

ptions :

406534365463. ✖

Recommended Daily Protein Intake

Gender:

Weight:

Recommended Intake: **grams**

406534365464. ✔

Recommended Daily Protein Intake

Gender: FEMALE

Weight: 60

Recommended Intake: **45 grams**

Recommended Daily Protein Intake

Official Answer Key: 6406534365464

MAD1_142 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 79 Question Id : 6406531527806 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535115551. ✖ http://127.0.0.1:5000/

406535115552. ✖ http://localhost:0000/

406535115553. ✖ http://0.0.0.0:5000/

406535115554. ✔ http://localhost:5008/

ub-Section Number : 3

Official Answer Key: 6406535115554

MAD1_153 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 90 Question Id : 6406531527818 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535115596. ✖ `<h1>Data added successfully</h1>`

406535115597. ✔ `<h1>Unsupported operation</h1>`

406535115598. ✖ `<h1>Method not Allowed</h1>`

406535115599. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535115597

MAD1_155 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 92 Question Id : 6406531527821 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

rom the given options, select which part of the code (if any) will throw an error first in the
erminal and prevent the Flask application from running successfully ?

ptions :

406535115604. ✔ `method="GET"` is not correct, it should be `methods=["GET"]`

406535115605. ✖ `def login(id)` can throw error because no `id` is passed in `/login` url

406535115606. ✖ `name` will throw error because no name will be written in the starting of app

406535115607. ✖ There is no error, Flask application will run correctly.

Official Answer Key: 6406535115604

MAD1_156 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 79 Question Id : 6406531528109 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535116394. ✖ http://127.0.0.1:5000/

406535116395. ✖ http://localhost:0000/

406535116396. ✖ http://0.0.0.0:5000/

406535116397. ✔ http://localhost:5008/

ub-Section Number : 3

Official Answer Key: 6406535116397

MAD1_166 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 90 Question Id : 6406531528121 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535116439. ✖ `<h1>Data added successfully</h1>`

406535116440. ✔ `<h1>Unsupported operation</h1>`

406535116441. ✖ `<h1>Method not Allowed</h1>`

406535116442. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535116440

MAD1_167 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 79 Question Id : 6406531528771 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535118325. ✖ http://127.0.0.1:5000/

406535118326. ✖ http://localhost:0000/

406535118327. ✖ http://0.0.0.0:5000/

406535118328. ✔ http://localhost:5008/

ub-Section Number : 3

Official Answer Key: 6406535118328

MAD1_177 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 90 Question Id : 6406531528783 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535118370. ✖ `<h1>Data added successfully</h1>`

406535118371. ✔ `<h1>Unsupported operation</h1>`

406535118372. ✖ `<h1>Method not Allowed</h1>`

406535118373. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535118371

MAD1_179 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 128 Question Id : 6406531533249 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535130945. ✖ http://127.0.0.1:5000/

406535130946. ✖ http://localhost:0000/

406535130947. ✖ http://0.0.0.0:5000/

406535130948. ✔ http://localhost:5008/

ub-Section Number :	3
ub-Section Id :	640653246672
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535130948

MAD1_189 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 139 Question Id : 6406531533261 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535130990. ✖ `<h1>Data added successfully</h1>`

406535130991. ✔ `<h1>Unsupported operation</h1>`

406535130992. ✖ `<h1>Method not Allowed</h1>`

406535130993. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535130991

MAD1_191 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 128 Question Id : 6406531533574 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535131841. ✖ http://127.0.0.1:5000/

406535131842. ✖ http://localhost:0000/

406535131843. ✖ http://0.0.0.0:5000/

406535131844. ✔ http://localhost:5008/

ub-Section Number :	3
ub-Section Id :	640653246768
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535131844

MAD1_201 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 139 Question Id : 6406531533586 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535131886. ✖ `<h1>Data added successfully</h1>`

406535131887. ✔ `<h1>Unsupported operation</h1>`

406535131888. ✖ `<h1>Method not Allowed</h1>`

406535131889. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535131887

MAD1_202 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 128 Question Id : 6406531533899 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535132737. ✖ http://127.0.0.1:5000/

406535132738. ✖ http://localhost:0000/

406535132739. ✖ http://0.0.0.0:5000/

406535132740. ✔ http://localhost:5008/

ub-Section Number :	3
ub-Section Id :	640653246864
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406535132740

MAD1_212 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 139 Question Id : 6406531533911 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535132782. ✖ `<h1>Data added successfully</h1>`

406535132783. ✔ `<h1>Unsupported operation</h1>`

406535132784. ✖ `<h1>Method not Allowed</h1>`

406535132785. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535132783

MAD1_213 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 65 Question Id : 6406531531440 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535125818. ✖ http://127.0.0.1:5000/

406535125819. ✖ http://localhost:0000/

406535125820. ✖ http://0.0.0.0:5000/

406535125821. ✔ http://localhost:5008/

ub-Section Number : 3

Official Answer Key: 6406535125821

MAD1_223 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 76 Question Id : 6406531531452 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535125863. ✖ `<h1>Data added successfully</h1>`

406535125864. ✔ `<h1>Unsupported operation</h1>`

406535125865. ✖ `<h1>Method not Allowed</h1>`

406535125866. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535125864

MAD1_225 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 65 Question Id : 6406531532323 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535128310. ✖ http://127.0.0.1:5000/

406535128311. ✖ http://localhost:0000/

406535128312. ✖ http://0.0.0.0:5000/

406535128313. ✔ http://localhost:5008/

ub-Section Number : 3

Official Answer Key: 6406535128313

MAD1_235 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 76 Question Id : 6406531532335 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535128355. ✖ `<h1>Data added successfully</h1>`

406535128356. ✔ `<h1>Unsupported operation</h1>`

406535128357. ✖ `<h1>Method not Allowed</h1>`

406535128358. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535128356

MAD1_236 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 65 Question Id : 6406531532886 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
Consider the following code:

app.py

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def port():
    return "This is the port question."

if __name__ == "__main__":
    app.run(port=5008)
```

ou are running a Flask web application locally using the command: `python app.py`. On which URL will the application be accessible on the browser?

ptions :

406535129880. ✖ http://127.0.0.1:5000/

406535129881. ✖ http://localhost:0000/

406535129882. ✖ http://0.0.0.0:5000/

406535129883. ✔ http://localhost:5008/

ub-Section Number : 3

Official Answer Key: 6406535129883

MAD1_246 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 76 Question Id : 6406531532898 Question Type : MCQ

orrect Marks : 4.5

uestion Label : Multiple Choice Question

f the given Flask app is run using the command

`ython server.py GET PUT POST` in one

erminal, what will be the output on another
erminal for the command

`curl -X POST http://127.0.0.1:5000/api/data`?

ptions :

406535129925. ✖ `<h1>Data added successfully</h1>`

406535129926. ✔ `<h1>Unsupported operation</h1>`

406535129927. ✖ `<h1>Method not Allowed</h1>`

406535129928. ✖ `<h1>404 Not Found</h1>`

Official Answer Key: 6406535129926

• 3.3 Routing, Dynamic URL Converters & View Functions (12 Questions)

MAD1_144 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 81 Question Id : 6406531527809 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following Flask view functions will return a 404 error for the URL:
`ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535115564. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535115565. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535115566. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535115567. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535115567

MAD1_157 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 81 Question Id : 6406531528112 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following Flask view functions will return a 404 error for the URL:
`ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535116407. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535116408. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535116409. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535116410. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535116410

MAD1_168 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 81 Question Id : 6406531528774 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following Flask view functions will return a 404 error for the URL:
`ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535118338. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535118339. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535118340. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535118341. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535118341

MAD1_180 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 130 Question Id : 6406531533252 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Which of the following Flask view functions will return a 404 error for the URL: `ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535130958. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535130959. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535130960. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535130961. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535130961

MAD1_192 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 130 Question Id : 6406531533577 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Which of the following Flask view functions will return a 404 error for the URL: `ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535131854. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535131855. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535131856. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535131857. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535131857

MAD1_203 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 130 Question Id : 6406531533902 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Which of the following Flask view functions will return a 404 error for the URL: `ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535132750. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535132751. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535132752. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535132753. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535132753

MAD1_214 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 67 Question Id : 6406531531443 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following Flask view functions will return a 404 error for the URL:
`ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535125831. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535125832. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535125833. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535125834. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535125834

MAD1_226 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 67 Question Id : 6406531532326 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following Flask view functions will return a 404 error for the URL:
`ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535128323. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535128324. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535128325. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535128326. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535128326

MAD1_237 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 67 Question Id : 6406531532889 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following Flask view functions will return a 404 error for the URL:
`ttp://127.0.0.1:5000/profile/user123/active/2024`?

ptions :

406535129893. ✖

```
@app.route('/profile/<username>/<status>/<int:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535129894. ✖

```
@app.route('/profile/<string:username>/<string:status>/<string:year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535129895. ✖

```
@app.route('/profile/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

406535129896. ✔

```
@app.route('/profile/user/<username>/<status>/<year>')
def profile(username, status, year):
    info = {'username': username, 'status': status, 'year': year}
    return info
```

Official Answer Key: 6406535129896

MAD1_267 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 108 Question Id : 640653995548 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following flask view functions will return a 404 error for the URL:
ttp://127.0.0.1:5000/communication/landline/0712 ?

ptions :

406533360945. ✖

```
app.route('/communication/<type>/<areacode>')
ef home(type, areacode):
   details = {'type': type, 'area-code': areacode}
   return details
```

406533360946. ✖

```
@app.route('/communication/<string:type>/<int:areacode>')
def home(type, areacode):
      details = {'type': type, 'area-code': areacode}
    return details
```

406533360947. ✖

```
@app.route('/communication/<string:type>/0712')
def home(type):
    details = {'type': type, 'area-code': 044}
    return details
```

406533360948. ✔

```
@app.route('/communication/online/<string:type>/<areacode>')
def home(type, areacode):
    details = {'type': type, 'area-code': 044}
    return details
```

ub-Section Number :	4
ub-Section Id :	640653146812
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533360948

MAD1_282 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 108 Question Id : 640653995830 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following flask view functions will return a 404 error for the URL:
ttp://127.0.0.1:5000/communication/landline/0712 ?

ptions :

406533361729. ✖

```
app.route('/communication/<type>/<areacode>')
ef home(type, areacode):
   details = {'type': type, 'area-code': areacode}
   return details
```

406533361730. ✖

```
@app.route('/communication/<string:type>/<int:areacode>')
def home(type, areacode):
      details = {'type': type, 'area-code': areacode}
    return details
```

406533361731. ✖

```
@app.route('/communication/<string:type>/0712')
def home(type):
    details = {'type': type, 'area-code': 044}
    return details
```

406533361732. ✔

```
@app.route('/communication/online/<string:type>/<areacode>')
def home(type, areacode):
    details = {'type': type, 'area-code': 044}
    return details
```

ub-Section Number :	4
ub-Section Id :	640653146901
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533361732

MAD1_295 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 108 Question Id : 640653996112 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
Which of the following flask view functions will return a 404 error for the URL:
ttp://127.0.0.1:5000/communication/landline/0712 ?

ptions :

406533362513. ✖

```
app.route('/communication/<type>/<areacode>')
ef home(type, areacode):
   details = {'type': type, 'area-code': areacode}
   return details
```

406533362514. ✖

```
@app.route('/communication/<string:type>/<int:areacode>')
def home(type, areacode):
      details = {'type': type, 'area-code': areacode}
    return details
```

406533362515. ✖

```
@app.route('/communication/<string:type>/0712')
def home(type):
    details = {'type': type, 'area-code': 044}
    return details
```

406533362516. ✔

```
@app.route('/communication/online/<string:type>/<areacode>')
def home(type, areacode):
    details = {'type': type, 'area-code': 044}
    return details
```

ub-Section Number :	4
ub-Section Id :	640653146990
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406533362516

• 3.5 Flask Request Object & Data Extraction (3 Questions)

MAD1_269 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 110 Question Id : 640653995541 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

onsider the following Python code snippet.

ilename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

f the above flask app is run using the command `python app.py GET POST` in one erminal, what will be the output on another terminal for command; url `http://127.0.0.1:5000/home` ?

ptions :

406533360916. ✗ `<h1>Hello from GET</h1>`

406533360917. ✓ `<h1>Hello from POST</h1>`

406533360918. ✗ `<h1>Please enter a valid HTTP method</h1>`

406533360919. ✗ `<h1>Method not Allowed</h1>`

Official Answer Key: 6406533360917

MAD1_283 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 110 Question Id : 640653995823 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

onsider the following Python code snippet.

ilename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

f the above flask app is run using the command `python app.py GET POST` in one erminal, what will be the output on another terminal for command; `url http://127.0.0.1:5000/home` ?

ptions :

406533361700. ✗ `<h1>Hello from GET</h1>`

406533361701. ✓ `<h1>Hello from POST</h1>`

406533361702. ✗ `<h1>Please enter a valid HTTP method</h1>`

406533361703. ✗ `<h1>Method not Allowed</h1>`

Official Answer Key: 6406533361701

MAD1_296 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 110 Question Id : 640653996105 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

onsider the following Python code snippet.

ilename: app.py

```
from flask import Flask, request
import sys
app = Flask(__name__)

@app.route('/home', methods = sys.argv)
def my_func():
    if request.method == 'GET':
        return "<h1>Hello from POST</h1>"

    elif request.method == 'POST':
        return "<h1>Hello from GET</h1>"

    else:
        return "<h1>Please enter a valid HTTP method<h1>"

app.run(debug = True)
```

f the above flask app is run using the command `python app.py GET POST` in one erminal, what will be the output on another terminal for command; `url http://127.0.0.1:5000/home` ?

ptions :

406533362484. ✗ `<h1>Hello from GET</h1>`

406533362485. ✓ `<h1>Hello from POST</h1>`

406533362486. ✗ `<h1>Please enter a valid HTTP method</h1>`

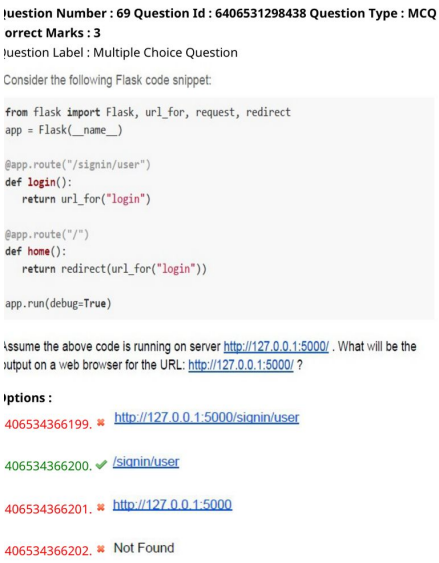
406533362487. ✗ `<h1>Method not Allowed</h1>`

Official Answer Key: 6406533362485

• 3.6 Flask Responses, Redirects & Error Handling (3 Questions)

MAD1_006 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf



uestion Number : 69 Question Id : 6406531298438 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Flask code snippet:

```
from flask import Flask, url_for, request, redirect
app = Flask(__name__)

@app.route("/signin/user")
def login():
   return url_for("login")

@app.route("/")
def home():
   return redirect(url_for("login"))

app.run(debug=True)
```

ssume the above code is running on server http://127.0.0.1:5000/ . What will be the utput on a web browser for the URL: http://127.0.0.1:5000/ ?

ptions :

406534366199. ✗ http://127.0.0.1:5000/signin/user

406534366200. ✓ /signin/user

406534366201. ✗ http://127.0.0.1:5000

406534366202. ✗ Not Found

Official Answer Key: 6406534366200

MAD1_099 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 69 Question Id : 6406531297898 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Flask code snippet:

```
from flask import Flask, url_for, request, redirect
app = Flask(__name__)

@app.route("/signin/user")
def login():
   return url_for("login")

@app.route("/")
def home():
   return redirect(url_for("login"))

app.run(debug=True)
```

ssume the above code is running on server http://127.0.0.1:5000/ . What will be the utput on a web browser for the URL: http://127.0.0.1:5000/ ?

ptions :

406534364657. ✖ http://127.0.0.1:5000/signin/user

406534364658. ✔ /signin/user

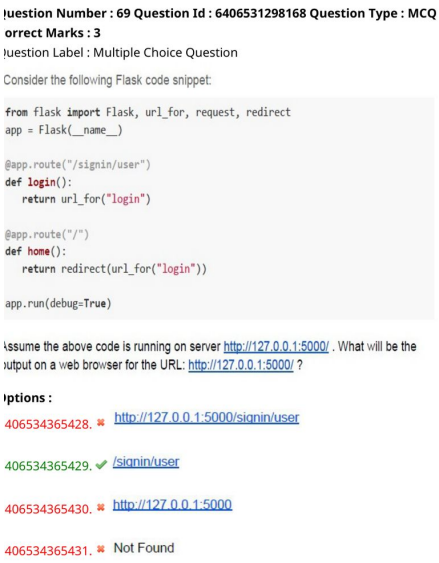
406534364659. ✖ http://127.0.0.1:5000

406534364660. ✖ Not Found

Official Answer Key: 6406534364658

MAD1_114 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf



uestion Number : 69 Question Id : 6406531298168 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Flask code snippet:

```
from flask import Flask, url_for, request, redirect
app = Flask(__name__)

@app.route("/signin/user")
def login():
   return url_for("login")

@app.route("/")
def home():
   return redirect(url_for("login"))

app.run(debug=True)
```

ssume the above code is running on server http://127.0.0.1:5000/ . What will be the utput on a web browser for the URL: http://127.0.0.1:5000/ ?

ptions :

406534365428. ✗ http://127.0.0.1:5000/signin/user

406534365429. ✓ /signin/user

406534365430. ✗ http://127.0.0.1:5000

406534365431. ✗ Not Found

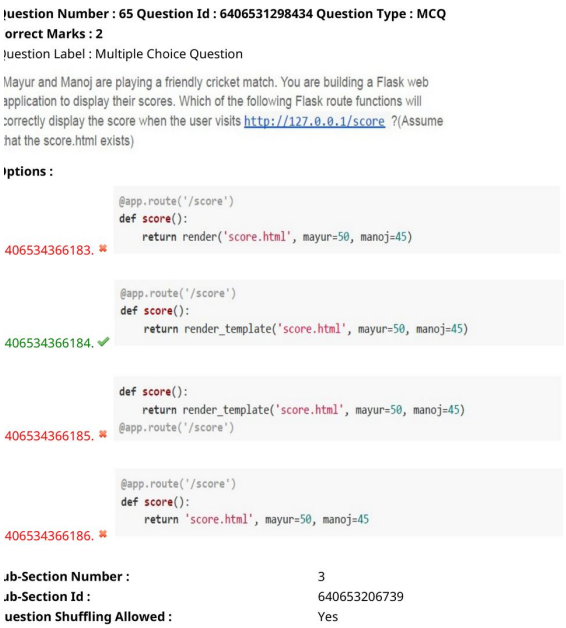
Official Answer Key: 6406534365429

WEEK 4: JINJA2 TEMPLATING, DYNAMIC RENDERING, SQLITE & CRUD APPLICATIONS

• 4.1 Jinja2 Templating Basics (Variables & Control Flow) (44 Questions)

MAD1_002 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf



uestion Number : 65 Question Id : 6406531298434 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

Mayur and Manoj are playing a friendly cricket match. You are building a Flask web application to display their scores. Which of the following Flask route functions will correctly display the score when the user visits http://127.0.0.1/score ?(Assume that the score.html exists)

ptions :

406534366183. ✗
```
@app.route('/score')
def score():
    return render('score.html', mayur=50, manoj=45)
```

406534366184. ✓
```
@app.route('/score')
def score():
    return render_template('score.html', mayur=50, manoj=45)
```

406534366185. ✗
```
def score():
    return render_template('score.html', mayur=50, manoj=45)
@app.route('/score')
```

406534366186. ✗
```
@app.route('/score')
def score():
    return 'score.html', mayur=50, manoj=45
```

ub-Section Number : 3
ub-Section Id : 640653206739
uestion Shuffling Allowed : Yes

Official Answer Key: 6406534366184

MAD1_004 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 67 Question Id : 6406531298436 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code snippet.

Filename: code.py

```
from string import Template as string_template
from jinja2 import Template as jinja_template

template1 = string_template('Hello $name, welcome to $city!')
template2 = jinja_template('{{product}} costs {{price}}')

result1 = template1.substitute(name='Alice')
result2 = template1.substitute(name='Alice', city='Boston',
state='Massachusetts')
result3 = template2.render({"product": "laptop"})
result4 = template2.render({"product": "laptop", "price": "45000",
"currency": "rupees"})

print(result1)
print(result2)
print(result3)
print(result4)
```

Which of the following statements will throw a key error?

ptions :

406534366191. ✔ `template1.substitute(name='Alice')`

406534366192. ✖ `template1.substitute(name='Alice', city='Boston', state='Massachusetts')`

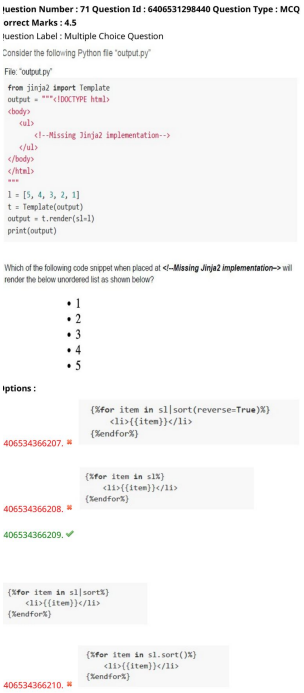
406534366193. ✖ `template2.render({"product": "laptop"})`

406534366194. ✖ `template2.render({"product": "laptop", "price": "45000", "currency": "rupees"})`

Official Answer Key: 6406534366191

MAD1_008 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf



uestion Number : 71 Question Id : 6406531298440 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following Python file "output.py"

File: "output.py"

```
from jinja2 import Template
output = """<!DOCTYPE html>
<body>
    <ul>
        <!--Missing Jinja2 implementation-->
    </ul>
</body>
</html>
"""
l = [5, 4, 3, 2, 1]
t = Template(output)
output = t.render(sl=l)
print(output)
```

Which of the following code snippet when placed at ***<!--Missing Jinja2 implementation-->*** will render the below unordered list as shown below?

- 1
- 2
- 3
- 4
- 5

ptions :

406534366207. ✖

```
{%for item in sl|sort(reverse=True)%}
    <li>{{item}}</li>
{%endfor%}
```

406534366208. ✖

```
{%for item in sl%}
    <li>{{item}}</li>
{%endfor%}
```

406534366209. ✔

```
{%for item in sl|sort%}
    <li>{{item}}</li>
{%endfor%}
```

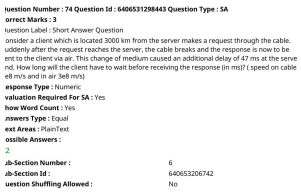
406534366210. ✖

```
{%for item in sl.sort()%}
    <li>{{item}}</li>
{%endfor%}
```

Official Answer Key: 6406534366209

MAD1_011 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf



uestion Number : 74 Question Id : 6406531298443 Question Type : SA
orrect Marks : 3
uestion Label : Short Answer Question
onsider a client which is located 3000 km from the server makes a request through the cable. uddenly after the request reaches the server, the cable breaks and the response is now to be ent to the client via air. This change of medium caused an additional delay of 47 ms at the serve nd. How long will the client have to wait before receiving the response (in ms)? (speed on cable e8 m/s and in air 3e8 m/s)
esponse Type : Numeric
valuation Required For SA : Yes
how Word Count : Yes
nswers Type : Equal
ext Areas : PlainText
ossible Answers :
2
ub-Section Number : 6
ub-Section Id : 640653206742
uestion Shuffling Allowed : No

uestion Id : 6406531298444 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix uestion Numbers : (75 to 77)
uestion Label : Comprehension
onsider the following Python code snippet and answer the given ubquestions
lename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'django': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }} framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]
    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language, lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

ub questions

Official Answer Key: 72

MAD1_051 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 73 Question Id : 6406531113765 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the template code and Python code given below:

emplate code: "output.html"

```
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
   <ol>
       {%for p in products%}
           {%if "i" in p.name|string %}
               <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
           {%endif%}
       {%endfor%}
   </ol>
</body>
</html>
```

ython code: "main.py"

```
from jinja2 import Template

products = [
   {"id": "101", "name": "Food", "qty": 78, "price": 100},
```

Official Answer Key: 6406533774165

MAD1_059 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf

uestion Number : 81 Question Id : 6406531113774 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

What will be the output of the given Python code snippet on the terminal for the command python app.py string ?

ptions :

406533774196. ✖
```
The book title is Python Basics
and it costs $20.
```

406533774197. ✔
```
The book title is $title and it
costs $20.
```

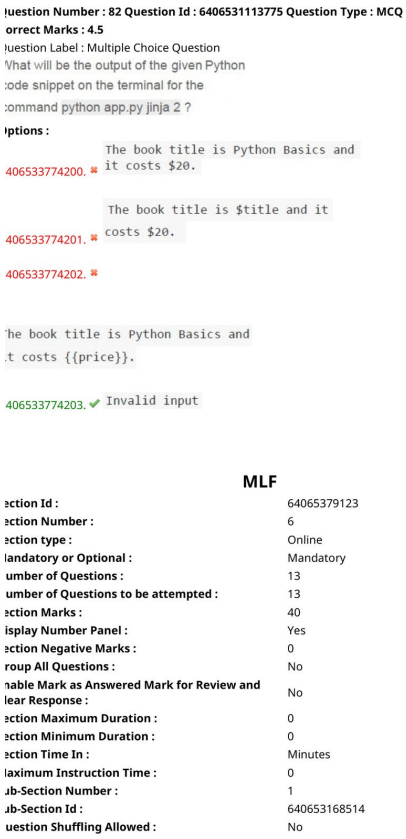
406533774198. ✖
```
The book title is Python Basics
and it costs {{price}}.
```

406533774199. ✖
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533774197

MAD1_060 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025.pdf



uestion Number : 82 Question Id : 6406531113775 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
What will be the output of the given Python
ode snippet on the terminal for the
ommand python app.py jinja 2 ?

ptions :

406533774200. ✗ `The book title is Python Basics and it costs $20.`

406533774201. ✗ `The book title is $title and it costs $20.`

406533774202. ✗

`he book title is Python Basics and t costs {{price}}.`

406533774203. ✓ `Invalid input`

MLF

ection Id :	64065379123
ection Number :	6
ection type :	Online
landatory or Optional :	Mandatory
umber of Questions :	13
umber of Questions to be attempted :	13
ection Marks :	40
isplay Number Panel :	Yes
ection Negative Marks :	0
roup All Questions :	No
nable Mark as Answered Mark for Review and lear Response :	No
ection Maximum Duration :	0
ection Minimum Duration :	0
ection Time In :	Minutes
laximum Instruction Time :	0
ub-Section Number :	1
ub-Section Id :	640653168514
uestion Shuffling Allowed :	No

Official Answer Key: 6406533774203

MAD1_065 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 73 Question Id : 6406531114392 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the template code and Python code given below:

emplate code: "output.html"

```
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
   <ol>
       {%for p in products%}
           {%if "i" in p.name|string %}
               <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
           {%endif%}
       {%endfor%}
   </ol>
</body>
</html>
```

ython code: "main.py"

```
from jinja2 import Template

products = [
   {"id": "101", "name": "Food", "qty": 78, "price": 100},
```

Official Answer Key: 6406533775978

MAD1_073 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 81 Question Id : 6406531114401 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

What will be the output of the given Python code snippet on the terminal for the command python app.py string ?

ptions :

406533776009. ✖
```
The book title is Python Basics
and it costs $20.
```

406533776010. ✔
```
The book title is $title and it
costs $20.
```

406533776011. ✖
```
The book title is Python Basics
and it costs {{price}}.
```

406533776012. ✖
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533776010

MAD1_074 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025.pdf

uestion Number : 82 Question Id : 6406531114402 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
What will be the output of the given Python
ode snippet on the terminal for the
ommand python app.py jinja 2 ?

ptions :

406533776013. ✗ The book title is Python Basics and it costs $20.

406533776014. ✗ The book title is $title and it costs $20.

406533776015. ✗

he book title is Python Basics and
t costs {{price}}.

406533776016. ✓ Invalid input

MLF

ection Id :	64065379156
ection Number :	6
ection type :	Online
landatory or Optional :	Mandatory
umber of Questions :	13
umber of Questions to be attempted :	13
ection Marks :	40
isplay Number Panel :	Yes
ection Negative Marks :	0
roup All Questions :	No
nable Mark as Answered Mark for Review and lear Response :	No
ection Maximum Duration :	0
ection Minimum Duration :	0
ection Time In :	Minutes
laximum Instruction Time :	0
ub-Section Number :	1
ub-Section Id :	640653168718
uestion Shuffling Allowed :	No

Official Answer Key: 6406533776016

MAD1_079 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 73 Question Id : 6406531114105 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

onsider the template code and Python code given below:

emplate code: "output.html"

```
<!DOCTYPE html>
<html lang="en">
<head>
</head>
<body>
   <ol>
       {%for p in products%}
           {%if "i" in p.name|string %}
               <li>{{p.id}}, {{p.name}}, {{p.qty}}, {{p.price}}</li>
           {%endif%}
       {%endfor%}
   </ol>
</body>
</html>
```

ython code: "main.py"

```
from jinja2 import Template

products = [
   {"id": "101", "name": "Food", "qty": 78, "price": 100},
```

Official Answer Key: 6406533775149

MAD1_087 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf

uestion Number : 81 Question Id : 6406531114114 Question Type : MCQ

orrect Marks : 3

uestion Label : Multiple Choice Question

What will be the output of the given Python code snippet on the terminal for the command python app.py string ?

ptions :

406533775180. ✖
```
The book title is Python Basics
and it costs $20.
```

406533775181. ✔
```
The book title is $title and it
costs $20.
```

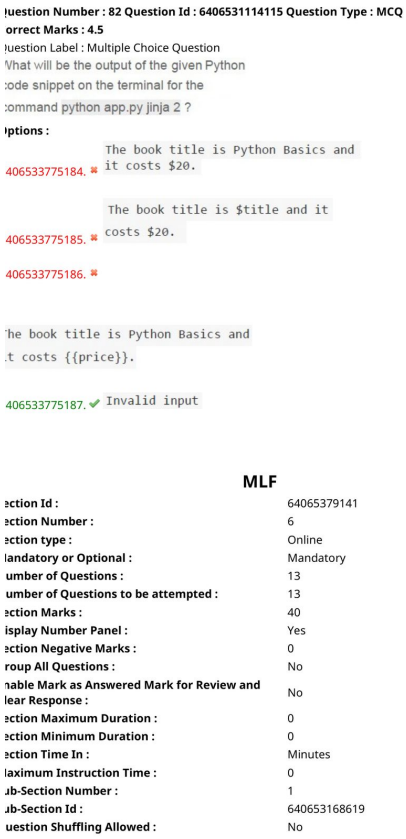
406533775182. ✖
```
The book title is Python Basics
and it costs {{price}}.
```

406533775183. ✖
```
The book title is $title and it
costs {{price}}.
```

Official Answer Key: 6406533775181

MAD1_088 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025.pdf



uestion Number : 82 Question Id : 6406531114115 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
What will be the output of the given Python
ode snippet on the terminal for the
ommand python app.py jinja 2 ?

ptions :

406533775184. ✗ The book title is Python Basics and it costs $20.

406533775185. ✗ The book title is $title and it costs $20.

406533775186. ✗

he book title is Python Basics and
t costs {{price}}.

406533775187. ✓ Invalid input

MLF

ection Id :	64065379141
ection Number :	6
ection type :	Online
landatory or Optional :	Mandatory
umber of Questions :	13
umber of Questions to be attempted :	13
ection Marks :	40
isplay Number Panel :	Yes
ection Negative Marks :	0
roup All Questions :	No
nable Mark as Answered Mark for Review and lear Response :	No
ection Maximum Duration :	0
ection Minimum Duration :	0
ection Time In :	Minutes
laximum Instruction Time :	0
ub-Section Number :	1
ub-Section Id :	640653168619
uestion Shuffling Allowed :	No

Official Answer Key: 6406533775187

MAD1_092 | Type: MCQ | Marks: +3.0

Source Exam(s): Quiz1 May23 .pdf

uestion Number : 109 Question Id : 640653577858 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

uestion Label : Multiple Choice Question

/hat will be the output of the following Python code snippet on the terminal.

```
rom string import Template

emp = "A student needs to complete $c1 level to get to the {{c2}}
evel."

emp = Template(temp)
ut = temp.substitute({'c1':'foundation', 'c2':'diploma'})
rint(out)
```

ptions :

406531929671. ✖

```
A student needs to complete $c1 level to get to the diploma
level.
```

406531929672. ✔

```
A student needs to complete foundation level to get to the
{{c2}} level.
```

406531929673. ✖

```
A student needs to complete foundation level to get to the
diploma level.
```

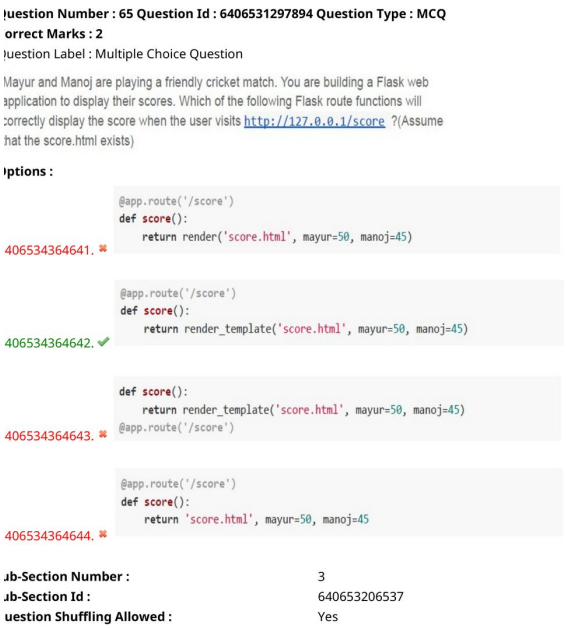
406531929674. ✖

```
KeyError: 'c2'
```

Official Answer Key: 6406531929672

MAD1_095 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf



uestion Number : 65 Question Id : 6406531297894 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

Mayur and Manoj are playing a friendly cricket match. You are building a Flask web application to display their scores. Which of the following Flask route functions will correctly display the score when the user visits http://127.0.0.1/score ?(Assume that the score.html exists)

ptions :

406534364641.
```
@app.route('/score')
def score():
    return render('score.html', mayur=50, manoj=45)
```

406534364642. ✔
```
@app.route('/score')
def score():
    return render_template('score.html', mayur=50, manoj=45)
```

406534364643.
```
def score():
    return render_template('score.html', mayur=50, manoj=45)
@app.route('/score')
```

406534364644.
```
@app.route('/score')
def score():
    return 'score.html', mayur=50, manoj=45
```

ub-Section Number :	3
ub-Section Id :	640653206537
uestion Shuffling Allowed :	Yes

Official Answer Key: 6406534364642

MAD1_097 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 67 Question Id : 6406531297896 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code snippet.

Filename: code.py

```
from string import Template as string_template
from jinja2 import Template as jinja_template

template1 = string_template('Hello $name, welcome to $city!')
template2 = jinja_template('{{product}} costs {{price}}')

result1 = template1.substitute(name='Alice')
result2 = template1.substitute(name='Alice', city='Boston',
state='Massachusetts')
result3 = template2.render({"product": "laptop"})
result4 = template2.render({"product": "laptop", "price": "45000",
"currency": "rupees"})

print(result1)
print(result2)
print(result3)
print(result4)
```

Which of the following statements will throw a key error?

ptions :

406534364649. ✔ `template1.substitute(name='Alice')`

406534364650. ✖ `template1.substitute(name='Alice', city='Boston', state='Massachusetts')`

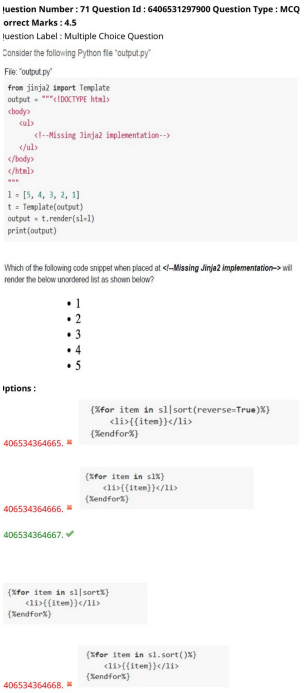
406534364651. ✖ `template2.render({"product": "laptop"})`

406534364652. ✖ `template2.render({"product": "laptop", "price": "45000", "currency": "rupees"})`

Official Answer Key: 6406534364649

MAD1_101 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf



uestion Number : 71 Question Id : 6406531297900 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following Python file "output.py"

File: "output.py"

```
from jinja2 import Template
output = """<!DOCTYPE html>
<body>
    <ul>
        <!--Missing Jinja2 implementation-->
    </ul>
</body>
</html>
"""
l = [5, 4, 3, 2, 1]
t = Template(output)
output = t.render(sl=l)
print(output)
```

Which of the following code snippet when placed at ***<!--Missing Jinja2 implementation-->*** will render the below unordered list as shown below?

- 1
- 2
- 3
- 4
- 5

ptions :

406534364665.
```
{%for item in sl|sort(reverse=True)%}
    <li>{{item}}</li>
{%endfor%}
```

406534364666.
```
{%for item in sl%}
    <li>{{item}}</li>
{%endfor%}
```

406534364667. ✔

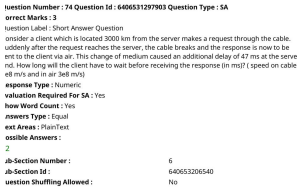
```
{%for item in sl|sort%}
    <li>{{item}}</li>
{%endfor%}
```

406534364668.
```
{%for item in sl.sort()%}
    <li>{{item}}</li>
{%endfor%}
```

Official Answer Key: 6406534364667

MAD1_103 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf



uestion Number : 74 Question Id : 6406531297903 Question Type : SA
orrect Marks : 3
uestion Label : Short Answer Question
onsider a client which is located 3000 km from the server makes a request through the cable. uddenly after the request reaches the server, the cable breaks and the response is now to be ent to the client via air. This change of medium caused an additional delay of 47 ms at the serve nd. How long will the client have to wait before receiving the response (in ms)? (speed on cable e8 m/s and in air 3e8 m/s)
esponse Type : Numeric
valuation Required For SA : Yes
how Word Count : Yes
nswers Type : Equal
ext Areas : PlainText
ossible Answers :
2
ub-Section Number : 6
ub-Section Id : 640653206540
uestion Shuffling Allowed : No

uestion Id : 6406531297904 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix uestion Numbers : (75 to 77)
uestion Label : Comprehension
onsider the following Python code snippet and answer the given ubquestions
filename: webapp.py

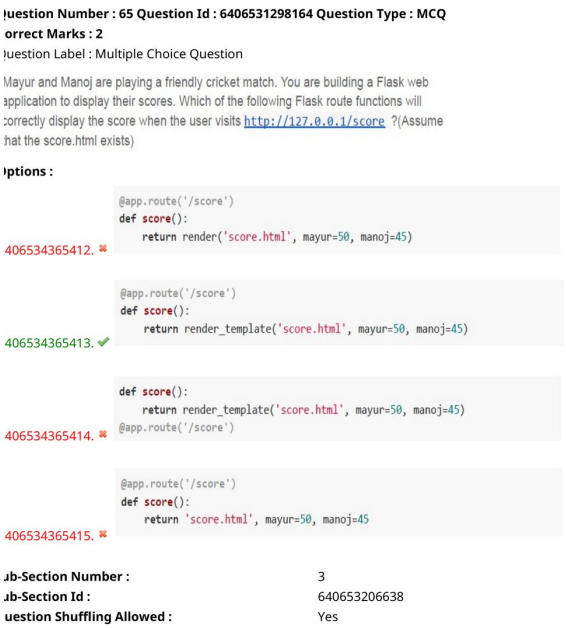
```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'django': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }} framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]
    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language, lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

ub questions

Official Answer Key: 72

MAD1_110 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf



uestion Number : 65 Question Id : 6406531298164 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question

Mayur and Manoj are playing a friendly cricket match. You are building a Flask web application to display their scores. Which of the following Flask route functions will correctly display the score when the user visits http://127.0.0.1/score ?(Assume that the score.html exists)

ptions :

406534365412. ✗

```
@app.route('/score')
def score():
    return render('score.html', mayur=50, manoj=45)
```

406534365413. ✓

```
@app.route('/score')
def score():
    return render_template('score.html', mayur=50, manoj=45)
```

406534365414. ✗

```
def score():
    return render_template('score.html', mayur=50, manoj=45)
@app.route('/score')
```

406534365415. ✗

```
@app.route('/score')
def score():
    return 'score.html', mayur=50, manoj=45
```

ub-Section Number : 3
ub-Section Id : 640653206638
uestion Shuffling Allowed : Yes

Official Answer Key: 6406534365413

MAD1_112 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 67 Question Id : 6406531298166 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code snippet.

Filename: code.py

```
from string import Template as string_template
from jinja2 import Template as jinja_template

template1 = string_template('Hello $name, welcome to $city!')
template2 = jinja_template('{{product}} costs {{price}}')

result1 = template1.substitute(name='Alice')
result2 = template1.substitute(name='Alice', city='Boston',
state='Massachusetts')
result3 = template2.render({"product": "laptop"})
result4 = template2.render({"product": "laptop", "price": "45000",
"currency": "rupees"})

print(result1)
print(result2)
print(result3)
print(result4)
```

Which of the following statements will throw a key error?

ptions :

406534365420. ✔ `template1.substitute(name='Alice')`

406534365421. ✖ `template1.substitute(name='Alice', city='Boston', state='Massachusetts')`

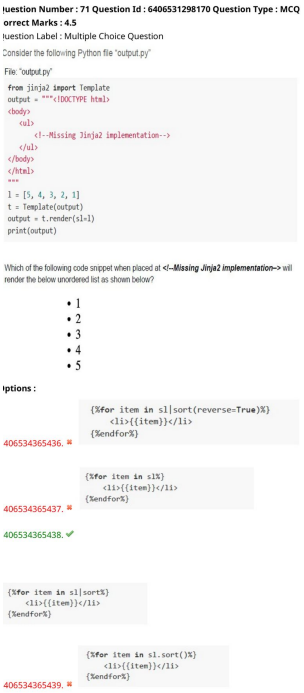
406534365422. ✖ `template2.render({"product": "laptop"})`

406534365423. ✖ `template2.render({"product": "laptop", "price": "45000", "currency": "rupees"})`

Official Answer Key: 6406534365420

MAD1_116 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf



uestion Number : 71 Question Id : 6406531298170 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following Python file "output.py"

File: "output.py"

```
from jinja2 import Template
output = """<!DOCTYPE html>
<body>
    <ul>
        <!--Missing Jinja2 implementation-->
    </ul>
</body>
</html>
"""
l = [5, 4, 3, 2, 1]
t = Template(output)
output = t.render(sl=l)
print(output)
```

Which of the following code snippet when placed at ***<!--Missing Jinja2 implementation-->*** will render the below unordered list as shown below?

- 1
- 2
- 3
- 4
- 5

ptions :

406534365436.
```
{%for item in sl|sort(reverse=True)%}
    <li>{{item}}</li>
{%endfor%}
```

406534365437.
```
{%for item in sl%}
    <li>{{item}}</li>
{%endfor%}
```

406534365438. ✔

```
{%for item in sl|sort%}
    <li>{{item}}</li>
{%endfor%}
```

406534365439.
```
{%for item in sl.sort()%}
    <li>{{item}}</li>
{%endfor%}
```

Official Answer Key: 6406534365438

MAD1_118 | Type: SA | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 74 Question Id : 6406531298173 Question Type : SA
orrect Marks : 3
uestion Label : Short Answer Question
onsider a client which is located 3000 km from the server makes a request through the cable. uddenly after the request reaches the server, the cable breaks and the response is now to be ent to the client via air. This change of medium caused an additional delay of 47 ms at the serve nd. How long will the client have to wait before receiving the response (in ms)? (speed on cable e8 m/s and in air 3e8 m/s)
esponse Type : Numeric
valuation Required For SA : Yes
how Word Count : Yes
nswers Type : Equal
ext Areas : PlainText
ossible Answers :
2
ub-Section Number : 6
ub-Section Id : 640653206641
uestion Shuffling Allowed : No

uestion Id : 6406531298174 Question Type : COMPREHENSION Sub Question Shuffling llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix uestion Numbers : (75 to 77)
uestion Label : Comprehension
onsider the following Python code snippet and answer the given ubquestions

filename: webapp.py

```
import sys
from jinja2 import Template

args = sys.argv
frameworks = {
    'django': 'python',
    'react': 'javascript',
    'flask': 'python',
    'vue': 'javascript'
}

languages = {
    'python': 'server-side',
    'javascript': 'client-side'
}

template = Template("{{ framework }} is a {{ lang_type }} {{ language }} framework.")

if len(args) >= 3 and args[1] in frameworks:
    framework_name = args[1]
    language = frameworks[framework_name]
    if args[2] == 'type' and language in languages:
        lang_type = languages[language]
        print(template.render(framework=framework_name, language=language, lang_type=lang_type))
    else:
        print("Invalid second argument or language not found!")
else:
    print("Invalid command")
```

ub questions

Official Answer Key: 72

MAD1_126 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 36 Question Id : 640653825028 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
/hat will be the output of the following Python code?

```
from jinja2 import Template
data = {"id": 101, "prod_name": "Groceries", "sales_amt": 5600,
"sales_bonus": 5}

def foo(amount, p):
   return amount * p / 100

a = foo(data["sales_amt"], data["sales_bonus"])
t = Template(
 "Congratulations, you got bonus {{ bonus }} by selling product
{{data['prod_name']}}"
)
output = t.render(data=data, bonus=a)
print(output)
```

ptions :

406532774759. ✖ `Congratulations, you got bonus 70.0 by selling product Groceries`

406532774760. ✖ `Congratulations, you got bonus 140.0 by selling product Groceries`

406532774761. ✔ `Congratulations, you got bonus 280.0 by selling product Groceries`

406532774762. ✖ `Congratulations, you got bonus 560.0 by selling product Groceries`

Official Answer Key: 6406532774761

MAD1_135 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 46 Question Id : 640653825035 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question
onsider the following flask app and the bodies of two index.html files shown in the figure below

app.py

```
from flask import Flask, render_template
import sys

app = ========== app object initialization here ===========

@app.route('/home')
def home():
    return render_template('index.html')

app.run()
```

app_templates/index.html

```
<h3>Home page</h3>
<p>MAD I focuses on backend development</p>
<h5>IITM ONLINE DEGREE</h5>
```

oot_templates/index.html

```
<h3>Home page</h3>
<p>Vue JS is a frontend technology</p>
<h5>IITM BS DEGREE</h5>
```

Official Answer Key: 6406532774790

MAD1_150 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 87 Question Id : 6406531527813 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535115582

MAD1_154 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf, IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 91 Question Id : 6406531527819 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
f the given Flask app is run using the command
`ython server.py GET PUT POST`
n one terminal, what will be the output on
nother terminal for the command
`url -X DELETE http://127.0.0.1:5000/api/data?`

ptions :

406535115600. ✗ `<h1>Data deleted successfully</h1>`

406535115601. ✗ `<h1>Unsupported operation</h1>`

406535115602. ✓ `<h1>Method not Allowed</h1>`

406535115603. ✗ `<h1>404 Not Found</h1>`

uestion Id : 6406531527820 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
uestion Numbers : (92 to 93)
uestion Label : Comprehension

Consider the following Flask code:

```python
from flask import Flask, render_template, request

app = Flask(__name__)

@app.route('/login', method="GET")
def login(id):
    name = request.form.get('name', 'Guest')
    return render_template("login.html", name=name)

if __name__ == '__main__':
    app.run(debug=True)
```

ogin.html

```html
<body>
    {{ name }}
    <form action="/login" method="POST">
        <input type="text" name="name" placeholder="Enter your name">
        <button type="submit">Submit</button>
    </form>
</body>
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535115602

MAD1_163 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 87 Question Id : 6406531528116 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535116425

MAD1_174 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 87 Question Id : 6406531528778 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535118356

MAD1_178 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf, IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 127 Question Id : 6406531533248 Question Type : MCQ

orrect Marks : 2

uestion Label : Multiple Choice Question

Which of the following is the correct syntax to display the total number of courses in a Jinja2 emplate?

ptions :

406535130941. ✖ `{{ courses | len }}`

406535130942. ✔ `{{ courses | length }}`

406535130943. ✖ `{{ len(courses) }}`

406535130944. ✖ None of these

Official Answer Key: 6406535130942

MAD1_186 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 136 Question Id : 6406531533256 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535130976

MAD1_190 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf, IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 140 Question Id : 6406531533262 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
f the given Flask app is run using the command
`ython server.py GET PUT POST`
n one terminal, what will be the output on
nother terminal for the command
`url -X DELETE http://127.0.0.1:5000/api/data?`

ptions :

406535130994. ✗ `<h1>Data deleted successfully</h1>`

406535130995. ✗ `<h1>Unsupported operation</h1>`

406535130996. ✓ `<h1>Method not Allowed</h1>`

406535130997. ✗ `<h1>404 Not Found</h1>`

uestion Id : 6406531533263 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
uestion Numbers : (141 to 142)
uestion Label : Comprehension

Consider the following Flask code:

```python
from flask import Flask, render_template, request

app = Flask(__name__)

@app.route('/login', method="GET")
def login(id):
    name = request.form.get('name', 'Guest')
    return render_template("login.html", name=name)

if __name__ == '__main__':
    app.run(debug=True)
```

ogin.html

```html
<body>
    {{ name }}
    <form action="/login" method="POST">
        <input type="text" name="name" placeholder="Enter your name">
        <button type="submit">Submit</button>
    </form>
</body>
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535130996

MAD1_198 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 136 Question Id : 6406531533581 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535131872

MAD1_209 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 136 Question Id : 6406531533906 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535132768

MAD1_220 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 73 Question Id : 6406531531447 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535125849

MAD1_224 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf, IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 77 Question Id : 6406531531453 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
f the given Flask app is run using the command
`ython server.py GET PUT POST`
n one terminal, what will be the output on
nother terminal for the command
`url -X DELETE http://127.0.0.1:5000/api/data?`

ptions :

406535125867. ✗ `<h1>Data deleted successfully</h1>`

406535125868. ✗ `<h1>Unsupported operation</h1>`

406535125869. ✓ `<h1>Method not Allowed</h1>`

406535125870. ✗ `<h1>404 Not Found</h1>`

uestion Id : 6406531531454 Question Type : COMPREHENSION Sub Question Shuffling
llowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
uestion Numbers : (78 to 79)
uestion Label : Comprehension

Consider the following Flask code:

```python
from flask import Flask, render_template, request

app = Flask(__name__)

@app.route('/login', method="GET")
def login(id):
    name = request.form.get('name', 'Guest')
    return render_template("login.html", name=name)

if __name__ == '__main__':
    app.run(debug=True)
```

ogin.html

```html
<body>
    {{ name }}
    <form action="/login" method="POST">
        <input type="text" name="name" placeholder="Enter your name">
        <button type="submit">Submit</button>
    </form>
</body>
```

ased on the above data, answer the given subquestions.
ub questions

Official Answer Key: 6406535125869

MAD1_232 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 73 Question Id : 6406531532330 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535128341

MAD1_243 | Type: MCQ | Marks: +4.5

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 73 Question Id : 6406531532893 Question Type : MCQ
orrect Marks : 4.5
uestion Label : Multiple Choice Question

Consider the following code:

```
from flask import Flask, request, render_template_string

app = Flask(__name__)

@app.route('/', methods=['GET', 'POST'])
def home():
    if request.method == 'POST':
        name = request.form['name']
        template = "Hello, {{ name }}!"
        return render_template_string(template, name=name)
    return '''
        <form method="POST">
            Enter Your Name: <input type="text" name="name">
            <button>Submit</button>
        </form>
    '''

if __name__ == '__main__':
    app.run(debug=True)
```

You are building a Flask web application that takes user input from an HTML form and displays

Official Answer Key: 6406535129911

MAD1_252 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 111 Question Id : 640653668557 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

uestion Label : Multiple Choice Question

Consider the following Python code snippet "code.py".

Filename: code.py

```
import sys
from jinja2 import Template

vars = sys.argv
myTech = {'mad 1': 'backend', 'mad 2': 'frontend'}

temp = Template("This course focuses on {{var}} development.")

if vars[1] == 'mad 1':
    print(temp.render(myTech['mad 1']))
elif vars[1] == 'mad 2':
    print(temp.render(myTech['mad 2']))
else:
    print("Mention the course name specifically.")
```

What will be the output on the terminal for input: `python code.py mad 1` ?

ptions :

406532239867. ✖ `This course focuses on frontend development.`

406532239868. ✖ `This course focuses on backend development.`

406532239869. ✔ `Mention the course name specifically.`

406532239870. ✖ `IndexError: list index out of range`

ub-Section Number : 4

Official Answer Key: 6406532239869

MAD1_263 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 104 Question Id : 640653995534 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following code block and select the correct option regarding the output when we run the python file app.py.
app.py

```
from jinja2 import Template

template='''
          Welcome to {Degree} DEGREE
          First level to clear is {{level}}
          '''

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

ptions :

406533360892. ✖ Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

406533360893. ✔ Welcome to {Degree} DEGREE
First level to clear is

406533360894. ✖ Welcome to IITM BS DEGREE
First level to clear is Foundation

406533360895. ✖ Welcome to DEGREE
First level to clear is {{ Level }}

Official Answer Key: 6406533360893

MAD1_266 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2.pdf

uestion Number : 107 Question Id : 640653995544 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

   def __init__(self, accno, loan_amount, interest_rate):
       self.accno = accno
       self.loan_amount = loan_amount
       self.interest_rate = interest_rate

   def get_loan_details(self):
       return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

ptions :

406533360929. ✖ **Account1 : (102, 170000, 6)**
Account2 : (101, 45900, 5)

406533360930. ✖ **Account2 : (102, 170000, 6)**

406533360931. ✖ **Account1 : (101, 45900, 5)**

406533360932. ✔ **Account1 : (101, 45900, 5)**
Account2 : (102, 170000, 6)

Official Answer Key: 6406533360932

MAD1_278 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 104 Question Id : 640653995816 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following code block and select the correct option regarding the output when we run the python file app.py.
app.py

```
from jinja2 import Template

template='''
          Welcome to {Degree} DEGREE
          First level to clear is {{level}}
          '''

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

ptions :

406533361676. ✖ Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

406533361677. ✔ Welcome to {Degree} DEGREE
First level to clear is

406533361678. ✖ Welcome to IITM BS DEGREE
First level to clear is Foundation

406533361679. ✖ Welcome to DEGREE
First level to clear is {{ Level }}

Official Answer Key: 6406533361677

MAD1_281 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3.pdf

uestion Number : 107 Question Id : 640653995826 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

   def __init__(self, accno, loan_amount, interest_rate):
       self.accno = accno
       self.loan_amount = loan_amount
       self.interest_rate = interest_rate

   def get_loan_details(self):
       return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

ptions :

406533361713. ✖ **Account1 : (102, 170000, 6)**
Account2 : (101, 45900, 5)

406533361714. ✖ **Account2 : (102, 170000, 6)**

406533361715. ✖ **Account1 : (101, 45900, 5)**

406533361716. ✔ **Account1 : (101, 45900, 5)**
Account2 : (102, 170000, 6)

Official Answer Key: 6406533361716

MAD1_291 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 104 Question Id : 640653996098 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following code block and select the correct option regarding the output when we run the python file app.py.
app.py

```
from jinja2 import Template

template='''
          Welcome to {Degree} DEGREE
          First level to clear is {{level}}
          '''

rendered=Template(template)
output=rendered.render(Degree="IITM BS",level="Foundation")
print(output)
```

ptions :

406533362460. ✖ Welcome to IITM BS DEGREE
First level to clear is {{ Level }}

406533362461. ✔ Welcome to {Degree} DEGREE
First level to clear is

406533362462. ✖ Welcome to IITM BS DEGREE
First level to clear is Foundation

406533362463. ✖ Welcome to DEGREE
First level to clear is {{ Level }}

Official Answer Key: 6406533362461

MAD1_294 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4.pdf

uestion Number : 107 Question Id : 640653996108 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Consider the following Python code and select correct output.

```
from jinja2 import Template

class Loan:

   def __init__(self, accno, loan_amount, interest_rate):
       self.accno = accno
       self.loan_amount = loan_amount
       self.interest_rate = interest_rate

   def get_loan_details(self):
       return (self.accno, self.loan_amount, self.interest_rate)

loan1 = Loan(101, 45900, 5)
loan2 = Loan(102, 170000, 6)

tm1 = Template("Account1 : {{l1.get_loan_details()}}")
tm2 = Template("Account2 : {{l2.get_loan_details()}}")

print(tm1.render(l1=loan1))
print(tm2.render(l2=loan2))
```

ptions :

406533362497. ✖ **Account1 : (102, 170000, 6)**
Account2 : (101, 45900, 5)

406533362498. ✖ **Account2 : (102, 170000, 6)**

406533362499. ✖ **Account1 : (101, 45900, 5)**

406533362500. ✔ **Account1 : (101, 45900, 5)**
Account2 : (102, 170000, 6)

Official Answer Key: 6406533362500

• 4.2 Jinja2 Filters, Macros & Template Inheritance (12 Questions)

MAD1_005 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 13 July 2025.pdf

uestion Number : 68 Question Id : 6406531298437 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
/hat will be the output of the following Jinja2 template when rendered in Flask?

```
{% macro greet(name) %}
Hello, {{ name }}!
{% endmacro %}

<!DOCTYPE html>
<html>
<head>
    <title>Jinja2 Macro Test</title>
</head>
<body>
    {{ greet("Alice") }}
    {{ greet("Bob") }}
</body>
</html>
```

ptions :

406534366195. ✔ Hello, Alice! Hello, Bob!

406534366196. ✖ `jinja2.exceptions.UndefinedError: 'greet' is undefined`

Hello, Alice!

406534366197. ✖ Hello, Bob!

{{ greet("Alice") }}

406534366198. ✖ {{ greet("Bob") }}

Official Answer Key: 6406534366195

MAD1_098 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 13 July 2025.pdf

uestion Number : 68 Question Id : 6406531297897 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
/hat will be the output of the following Jinja2 template when rendered in Flask?

```
{% macro greet(name) %}
Hello, {{ name }}!
{% endmacro %}

<!DOCTYPE html>
<html>
<head>
    <title>Jinja2 Macro Test</title>
</head>
<body>
    {{ greet("Alice") }}
    {{ greet("Bob") }}
</body>
</html>
```

ptions :

406534364653. ✔ Hello, Alice! Hello, Bob!

406534364654. ✖ `jinja2.exceptions.UndefinedError: 'greet' is undefined`

406534364655. ✖ Hello, Alice!
Hello, Bob!

406534364656. ✖ {{ greet("Alice") }}
{{ greet("Bob") }}

Official Answer Key: 6406534364653

MAD1_113 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 13 July 2025.pdf

uestion Number : 68 Question Id : 6406531298167 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question
/hat will be the output of the following Jinja2 template when rendered in Flask?

```
{% macro greet(name) %}
Hello, {{ name }}!
{% endmacro %}

<!DOCTYPE html>
<html>
<head>
    <title>Jinja2 Macro Test</title>
</head>
<body>
    {{ greet("Alice") }}
    {{ greet("Bob") }}
</body>
</html>
```

ptions :

406534365424. ✔ Hello, Alice! Hello, Bob!

406534365425. ✖ `jinja2.exceptions.UndefinedError: 'greet' is undefined`

406534365426. ✖ Hello, Alice!
Hello, Bob!

406534365427. ✖ {{ greet("Alice") }}
{{ greet("Bob") }}

Official Answer Key: 6406534365424

MAD1_146 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025.pdf

uestion Number : 83 Question Id : 6406531527811 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535115572. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535115573. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535115574. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535115575. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4

ub-Section Id : 640653244910

uestion Shuffling Allowed : Yes

Official Answer Key: 6406535115574

MAD1_159 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025.pdf

uestion Number : 83 Question Id : 6406531528114 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535116415. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535116416. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535116417. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535116418. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4

ub-Section Id : 640653245006

uestion Shuffling Allowed : Yes

Official Answer Key: 6406535116417

MAD1_170 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025.pdf

uestion Number : 83 Question Id : 6406531528776 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535118346. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535118347. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535118348. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535118349. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4

ub-Section Id : 640653245191

uestion Shuffling Allowed : Yes

Official Answer Key: 6406535118348

MAD1_182 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID2 26 Oct 2025.pdf

uestion Number : 132 Question Id : 6406531533254 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535130966. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535130967. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535130968. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535130969. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4

ub-Section Id : 640653246673

uestion Shuffling Allowed : Yes

Official Answer Key: 6406535130968

MAD1_194 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID3 26 Oct 2025.pdf

uestion Number : 132 Question Id : 6406531533579 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535131862. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535131863. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535131864. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535131865. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4
ub-Section Id : 640653246769
uestion Shuffling Allowed : Yes

Official Answer Key: 6406535131864

MAD1_205 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QID4 26 Oct 2025.pdf

uestion Number : 132 Question Id : 6406531533904 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535132758. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535132759. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535132760. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535132761. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4
ub-Section Id : 640653246865
uestion Shuffling Allowed : Yes

Official Answer Key: 6406535132760

MAD1_216 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT2 26 Oct 2025.pdf

uestion Number : 69 Question Id : 6406531531445 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535125839. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535125840. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535125841. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535125842. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4
ub-Section Id : 640653246064
uestion Shuffling Allowed : Yes

Official Answer Key: 6406535125841

MAD1_228 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT3 26 Oct 2025.pdf

uestion Number : 69 Question Id : 6406531532328 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535128331. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535128332. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535128333. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535128334. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4

ub-Section Id : 640653246357

uestion Shuffling Allowed : Yes

Official Answer Key: 6406535128333

MAD1_239 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M IMPROVEMENT AN EXAM QIT4 26 Oct 2025.pdf

uestion Number : 69 Question Id : 6406531532891 Question Type : MCQ
orrect Marks : 3
uestion Label : Multiple Choice Question

Match the following web development concepts with their correct descriptions:

Column A	Column B
1. PyHTML	A. Lightweight engine in Python for embedding variables inside strings using `$variable` style
2. Jinja2	B. A Python library to build HTML structures programmatically using Python functions
3. Template Rendering	C. The process of generating dynamic HTML by combining templates with data
4. String Template Rendering	D. Templating engine used by Flask that supports control structures, filters, and inheritance

ptions :

406535129901. ✖ 1 → B, 2 → C, 3 → D, 4 → A

406535129902. ✖ 1 → A, 2 → B, 3 → C, 4 → D

406535129903. ✔ 1 → B, 2 → D, 3 → C, 4 → A

406535129904. ✖ 1 → C, 2 → D, 3 → A, 4 → B

ub-Section Number : 4
ub-Section Id : 640653246552
uestion Shuffling Allowed : Yes

Official Answer Key: 6406535129903

• 4.4 Database Integration in Flask (sqlite3, Cursor & Connection) (1 Questions)

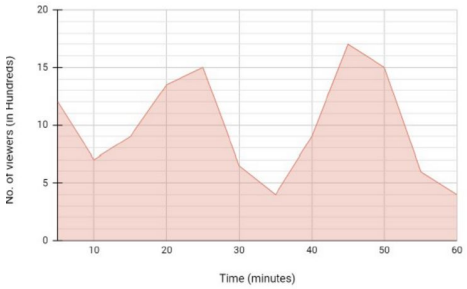
MAD1_251 | Type: MCQ | Marks: +3.0

Source Exam(s): IIT M DIPLOMA AN2 EXAM QPD2 29 Oct.pdf, IIT M DIPLOMA AN3 EXAM QPD3 29 Oct.pdf

uestion Number : 109 Question Id : 640653668555 Question Type : MCQ Is Question
landatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instructi
ime : 0
orrect Marks : 3

uestion Label : Multiple Choice Question

certain video that was available on the web only for an hour occupied 2 gigabytes of memory c
ne server. If the number of viewers of the video over time for that hour is given by the curve
hown below and assuming that each viewer required an individual connection to the server to
iew the video, what would be the minimum RAM requirement of the server that could process
equests from all the viewers simultaneously at any point of time?



ptions :

406532239859. ✖ 1.7 GB

406532239860. ✖ 3.4 GB

406532239861. ✖ 1.7 TB

406532239862. ✔ 3.4 TB

Official Answer Key: 6406532239862

• 4.5 Full-Stack CRUD Workflows & Form Processing (1 Questions)

MAD1_127 | Type: MCQ | Marks: +2.0

Source Exam(s): IIT M DIPLOMA AN EXAM QDD2 7 July 2024.pdf, IIT M DIPLOMA AN EXAM QDD3 7 July 2024.pdf

uestion Number : 37 Question Id : 640653825034 Question Type : MCQ
orrect Marks : 2
uestion Label : Multiple Choice Question
onsider the below table.

Column 1	Column 2
) Primary Key	a) Method of the request
i) Foreign key	b) Provides additional functionality at top of original function
ii) Nullable	c) Refers to another parent table column
v) POST	d) Unique identification of row
) @<decorator>	e) Undefined value

se the above table. Which of the below match is correct?
ptions :
406532774783. ✖ i - d, ii - b, iii - a, iv - e, v - c
406532774784. ✔ i - d, ii - c, iii - e, iv - a, v - b
406532774785. ✖ i - c, ii - d, iv - e, iii - a, v - b
406532774786. ✖ i - c, ii - d, iv - e, iii - b, v - a

ub-Section Number : 3
ub-Section Id : 640653122721
uestion Shuffling Allowed : Yes

Official Answer Key: 6406532774784